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Laboratory observations of a transient turbulent Ekman layer

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We investigate the transient dynamics of a turbulent Ekman boundary layer through large-scale spin-up experiments on the Coriolis rotating platform in Grenoble. An impulsive change in rotation rate generates a well-controlled bottom stress, exciting inertial oscillations that decay over several inertial periods, more slowly than laminar theory predicts. We show that this behaviour is controlled by the bottom stress formulation: a quadratic drag law $\tau_b = C_f|\mathbf{u}|\mathbf{u}$, with $C_f \approx 5 \times 10^{-3}$, correctly reproduces both the damping rate and the interior spin-up. Despite the slow attenuation, the steady Ekman spiral and the inertial oscillations remain consistent with an approximately linear superposition, extending a result trivially expected in the laminar case to the turbulent regime. The k-ε closure appears to be in good agreement with the data.

Keywords: Turbulent Ekman layer, Inertial oscillations, Spin-up, Bottom drag, k-epsilon model

1. Introduction

Oceanic boundary layers at the surface and seafloor are key sites of energy injection and dissipation (Wunsch and Ferrari 2004). At the surface, wind stress drives a turbulent mixed layer whose depth controls air–sea exchanges of heat, gases, and momentum. At the seafloor, bottom friction dissipates the kinetic energy of geostrophic currents and tidal flows, and sets the bottom boundary condition for the large-scale circulation (Pedlosky 1996, Vallis 2006). Yet, representing these layers in numerical models remains a challenge, as their turbulent structure depends sensitively on the choice of closure scheme and boundary conditions.

Observations of the bottom boundary layer over continental shelves and in the deep ocean have documented the Ekman spiral, the log-layer structure of the mean velocity, and the inhibition of turbulence by stable density stratification (Weatherly and Martin 1978, Perlin et al. 2005, 2007). At the ocean surface, direct current measurements have also confirmed the Ekman spiral and the wind-driven mixed layer deepening (McPhaden et al. 2024, Price et al. 1978). Laboratory experiments have played a complementary role in advancing our understanding of turbulent boundary layers. In particular, the spin-up experiment of Kato and Phillips (1969), in which a stress is applied at the surface of a stratified fluid, has become a standard benchmark for testing turbulence parameterizations (Price et al. 1986, Umlauf and Burchard 2003).

The dynamical regime in the boundary layer under the influence of constant surface stress has been described in the seminal work of Ekman (1905), who showed that the steady balance between the Coriolis force and the divergence of the vertical momentum flux gives rise to a

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veering spiral confined within a depth $\delta_E = \sqrt{2\nu/f}$, where f denotes the Coriolis parameter and ν the kinematic viscosity.

In geophysical flows, molecular viscosity can be tentatively replaced by a turbulent eddy viscosity $\nu_t \sim u_* \delta_E$ (with u_* a friction velocity), yielding the turbulent scaling $\delta_E \propto u_*/f$ (Rossby and Montgomery 1935, Zilitinkevich and Mironov 1996). Unlike the laminar case, for which the momentum equations are linear and admit an exact analytical solution, turbulence introduces nonlinearities through the Reynolds stress terms, which couple the mean flow to the velocity fluctuations. The eddy viscosity $\nu_t(z,t)$ is no longer a known quantity but must be modeled, giving rise to the turbulence closure problem.

Several modelling approaches have been proposed to better understand the evolution of the mixed layer subject to an interfacial stress, as reviewed by Kim et al. (2014): slab models, which ignore the vertical structure entirely (Pollard and Millard 1970, D'Asaro 1985, Alford 2001); imposed eddy-viscosity profiles $\nu_t(z)$, which prescribe the turbulent mixing without a dynamical equation for the turbulent kinetic energy (Faller and Kaylor 1966, Madsen 1977, Endoh and Nitta 1971, Lewis and Belcher 2004). Another approach is to use linear Rayleigh damping $-r\mathbf{u}$, where the empirical coefficient r aggregates unresolved mechanisms such as wave breaking and turbulent dissipation (Csanady and Shaw 1980, Ashkenazy et al. 2015, Kim et al. 2014). In coarse-resolution ocean models, linear or quadratic friction coefficients are typically defined as a bulk drag coefficient relating the bottom stress to the depth-averaged or near-bottom velocity. More sophisticated two-equation closures, such as the k-ε model, have been applied to the steady turbulent Ekman layer and shown to agree well with high-resolution DNS (Braun et al. 2020, Marlatt et al. 2012).

However, the steady turbulent Ekman layer is only part of the picture. Wind stress is inherently unsteady. Impulsive or intermittent forcing excites inertial oscillations which carry a significant fraction of the near-surface kinetic energy and represent a major pathway for momentum transfer into the deep ocean (Pollard and Millard 1970, D'Asaro 1985, Alford 2001). In the linear framework, the transient inertial response and the steady Ekman spiral simply superpose (Pollard and Millard 1970): inertial oscillations produced by the onset of a constant stress decay through frictional dissipation, leaving eventually the steady Ekman balance. In the turbulent regime, however, this decoupling breaks down: the oscillating shear modulates the eddy viscosity $\nu_t(z,t)$, which in turn alters the vertical Ekman spiral profile (Coppin et al. 2025). This two-way coupling is a fundamentally nonlinear effect, and its consequences for the decay rate of inertial oscillations and the structure of the turbulent Ekman layer remain poorly documented.

One goal of this study is to verify whether the laminar solution can be extended to turbulent regimes. We also would like to assess the sensitivity of inertial oscillation decay to the boundary conditions. Last, we would like to validate the k-ε turbulent closure in this transient, rotation-dominated regimes.

To address these questions, we present large-scale spin-up experiments conducted on the Coriolis rotating platform in Grenoble (France). The spin-up process has been extensively studied since Greenspan and Howard (1963) (see the review by Duck and Foster (2001)), but few experiments have reached turbulent regimes (Caldwell et al. 1972, Sous et al. 2013), and none have documented the transient oscillating stage that immediately follows the impulsive forcing. The large Reynolds numbers achieved on the Coriolis platform allow the boundary layer to be fully turbulent.We focus on the weakly stratified limit, for which rotation rather than buoyancy constrains the vertical structure: in strongly stratified regimes the Ekman layer becomes buoyancy-limited to a depth $\delta_E\sqrt{f/N}$ (Pollard et al. 1973, Fernando 1991, Kato and Phillips 1969). The threshold value of f/N at which this transition occurs remains poorly constrained (Zilitinkevich 2012). We investigate the regime $f/N \sim \mathcal{O}(1)$, where rotation rather than buoyancy is expected to control the vertical structure of the flow.

The paper is organized as follows. The experimental setup and measurement techniques are

described in Section 2. Section 3 develops the theoretical framework: the 1D axisymmetric model, the friction law, and inertial oscillations together with laminar analytical solutions for the no-slip and free-slip limits. Results are presented in Section 4 and Section 5, where mean flow profiles, turbulent statistics, and the transient inertial response are compared against the k-ε closure. Section 6 summarizes the findings and discusses their implications for the parameterization of oceanic boundary layers.

2. Experimental Methods

To investigate the transient dynamics of a turbulent Ekman layer under controlled conditions, laboratory experiments were carried out in the Coriolis Platform at Grenoble (Fig. 1). The circular tank (13 m in diameter) is filled with water to a depth $H = 0.5$ m. The forcing is produced by an impulsive change in the platform rotation speed, which generates an azimuthal flow in the platform frame of reference. This procedure is referred to as *spin-up* when the rotation speed increases, and *spin-down* when it decreases. The platform reaches its nominal rotation speed after an acceleration phase of 20 s.

Throughout this paper, we use cylindrical coordinates (e_r, e_θ, e_z) when describing the flow in the tank. The vertical coordinate z is oriented upward, with origin $z = 0$ at the tank floor (where the forcing is applied). The three velocity components (u_r, u_θ, u_z) are measured by Stereoscopic Particle Image Velocimetry (S-PIV) in a vertical plane tangent to the azimuthal direction at the fixed radius $r_0 = 4.5$ m.

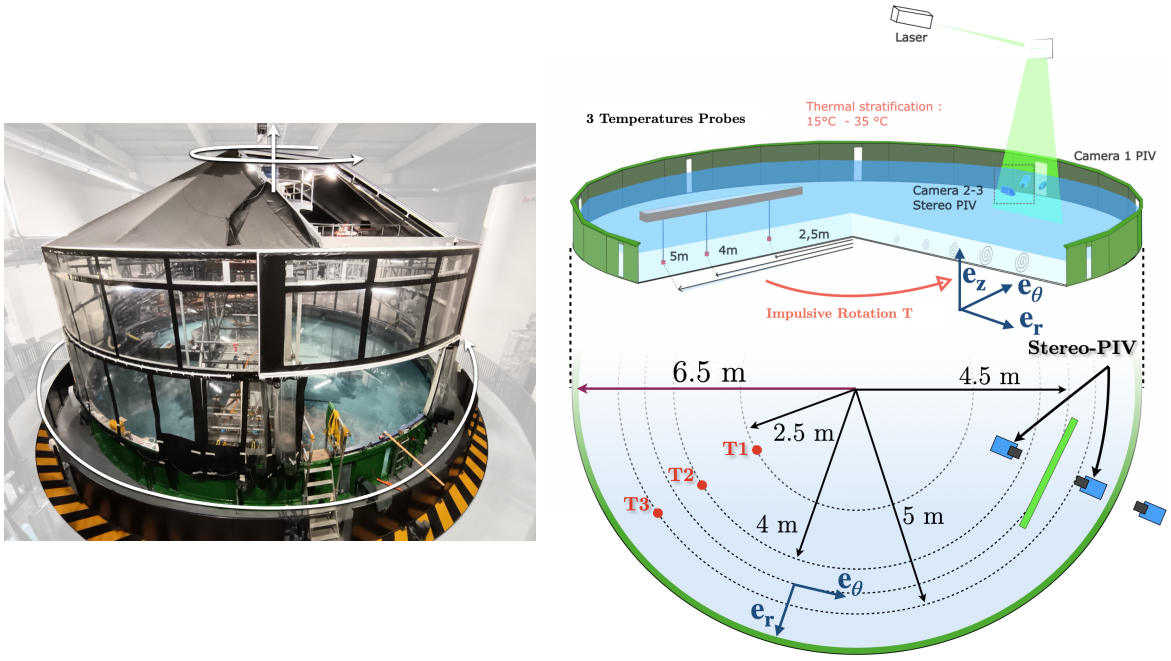


Figure 1. Illustration of the experimental apparatus. Left: Picture of the 13 m wide rotating Coriolis Platform. Right: Schematic side and top view of the experimental set-up. The water height in the tank is 50 cm.

2.1. *Experimental Configuration*

The experiments are characterized by three main external control parameters: the Coriolis parameter $f = 2\Omega_i$, set by the initial background rotation rate Ω_i of the platform, the im-

pulsive change in rotation speed $\Delta\Omega = \Omega_f - \Omega_i$, and the buoyancy frequency N, set by the initial density stratification.

2.1.1. *Background rotation*

The platform rotates at an initial angular velocity Ω_i, which sets the background rotation of the fluid and defines the Coriolis parameter $f = 2\Omega_i$. In the rotating experiments, $\Omega_i = 0.0524$ rad s^{-1}, corresponding to an inertial period $T_f = 2\pi/f \approx 63$ s. In the non-rotating reference case, $\Omega_i = 0$ and the relevant time scale is instead set by the impulsive forcing $\Delta\Omega$.

2.1.2. *Mechanical Forcing*

The impulsive change in the platform rotation rate appears as a suddenly imposed bottom velocity in the initial reference frame, while the fluid far above remains at rest temporarily due to inertia. In all the experiments presented here, the same rotation rate increment $\Delta\Omega = 0.0524$ rad s^{-1} is applied. At the measurement radius $r_0 = 4.5$ m, it results in a given bottom velocity $v_0 = r_0\ \Delta\Omega = 23.6$ cm s^{-1}. The resulting bottom stress $\tau = u_*^2$, where u_* is the friction velocity, drives the formation of the boundary layer and the subsequent spin-up of the fluid column. Due to the veering effect of the Coriolis force, the stress is tilted by an angle α_0 with respect to the azimuthal direction, so we must distinguish the azimuthal shear stress $\tau_\theta = u_*^2 \cos\alpha_0$ and the radial shear stress $\tau_r = -u_*^2 \sin\alpha_0$.

The friction velocity will be estimated from the measurements of the total vertical flux of azimuthal momentum,

$$\tau_\theta = \langle v'w' \rangle\big|_{z\to 0} - \nu \left.\frac{\partial \langle v \rangle}{\partial z}\right|_{z\to 0}, \quad (1)$$

which remains approximately constant with z in the wall layer. Since the spatial resolution of the S-PIV does not allow us to resolve the viscous sublayer, u_* is estimated by extrapolating the measured momentum flux profile toward the wall. We can similarly estimate the radial stress τ_r from the radial velocity component u. Typical values obtained for the friction velocity are $u_* \simeq 1$ cm s^{-1} during the initial phase of motion (see section 4.2). Note however that u_* slowly decreases during the spin-up as the relative velocity between the plate and the interior flow decays.

2.1.3. *Density stratification*

We perform experiments with no stratification to study the canonical Ekman layer. Along with these experiments, we consider the weakly stratified limit to provide a comparison with the reference experiment of Kato and Phillips (1969). This stable density stratification is due to a constant vertical temperature gradient established prior to each experiment by filling the tank progressively from the bottom. The injected water is a mixture of water preheated to 40°C and cold tap water at approximately 17°C, with the hot fraction decreasing continuously from 100% at the start to 0% at the end of the filling. To avoid turbulent mixing during filling while limiting heat loss to the atmosphere, the tank is filled over 4 h to reach a depth of $H = 50$ cm. The resulting density profile is characterized by the buoyancy frequency N defined by

$$N^2 = -\frac{g}{\rho_0}\frac{\partial \rho}{\partial z} = g\alpha_T \frac{\partial T}{\partial z}, \quad (2)$$

where α_T is the thermal expansion coefficient and ρ_0 a mean reference density. The value of N is approximately uniform over the lower 30 cm of the water column. The upper layer (≈ 20 cm) is mixed due to convection at the water–air interface; we do not consider this upper layer further in the analysis. The stratification obtained with temperature is weaker than those usually obtained with salinity, enabling us to explore the limit of weak stratification. The choice of a temperature stratification instead of the more usual salt stratification was motivated as part of a wider project involving thermal convection, not discussed here.

2.1.4. *Overview of the experiments*

In this article, we analyse four main experiments defined by the presence or absence of background rotation and stable stratification, as summarized in Table 1.

Label	f (rad s^{-1})	N (s^{-1})
EXP-Ref	0	0
EXP-S	0	0.25
EXP-R	0.105	0
EXP-SR	0.105	0.32

Table 1. Experiment names with values of the control parameters. f is the Coriolis parameter, N the buoyancy frequency. In all the experiments, the water depth is $H = 50$ cm and $\Delta\Omega = 0.0524$ rad s^{-1}, corresponding to a bottom velocity $v_0 = 23.5$ cm s^{-1} at the reference radius $r_0 = 4.5$ m, where velocity measurements are made.

2.2. *Measurement techniques*

We use three complementary techniques to observe the flow: laser-induced fluorescence (LIF), which gives a qualitative picture of the flow structure; Stereoscopic PIV (S-PIV), which provides the three velocity components in a vertical plane; and vertically profiling thermistor probes, which resolve the temperature stratification.

2.2.1. *Laser-induced fluorescence*

Figure 2 presents a snapshot of the flow during the transient growth of the boundary layer, visualised using fluorescent dye (rhodamine 6G) injected at the tank floor at the beginning of the experiment. The image is taken in a plane tangent to the cylinder at radius $r_0 = 4.5$ m, which locally approximates the $(\mathbf{e}_\theta, \mathbf{e}_z)$ surface since the visualization width $L_{\rm laser} = 32$ cm is small compared with r_0. Immediately above the floor lies the viscous sublayer, a thin region where viscous stresses dominate and the velocity profile is linear, which is too thin to be visible in the figure (a thickness about 0.5 mm is expected). Above it lies a turbulent wall layer where the total shear stress remains nearly equal to its wall value u_*^2. We shall call it the log layer, although the theoretical logarithmic velocity profile is only marginally reached in our range of Reynolds number. Above this log layer, the mixed layer is characterised by an approximately uniform dye concentration, reflecting efficient scalar mixing by turbulent eddies. The velocity field within this layer, however, retains significant vertical structure: it decreases monotonically with height in the non-rotating case, and adopts an Ekman-like profile when background rotation is present. In the rotating case, the Coriolis force confines this active layer to a depth $\delta_E \sim u_*/f$, beyond which the flow decouples from the bottom stress. Note that this situation of weak stratification differs from the case of strong stratification $N/f \gg 1$, for which the mixed layer is limited by stratification to a thickness smaller than δ_E.

At the top of the mixed layer, the entrainment layer marks the transition to the quiescent stratified interior above. It is visible in Fig. 2 as the bright, turbulent interface where overturning structures erode the stratification. The entrainment is driven by Kelvin–Helmholtz billows, which arise when the local Richardson number falls below $Ri_c \approx 0.25$ (Miles 1961, Howard 1961). Overturning intrusions and small-scale eddies then engulf pockets of stratified fluid into the mixed layer.

2.2.2. *Stereoscopic Particle Image Velocimetry (S-PIV) and image processing*

The velocity field is also measured in a vertical plane located at $r_0 = 4.5$ m from the centre (2 m from the outer boundary) using Stereoscopic Particle Image Velocimetry (S-PIV). The fluid is seeded with polyamide particles (Orgasol, 60 μm diameter, density 1.02), which are approximately neutrally buoyant (the settling velocity computed from Stokes law is $w_s \sim 4 \times 10^{-3}$ cm s^{-1}, negligible compared to the flow velocity). The particles are illuminated

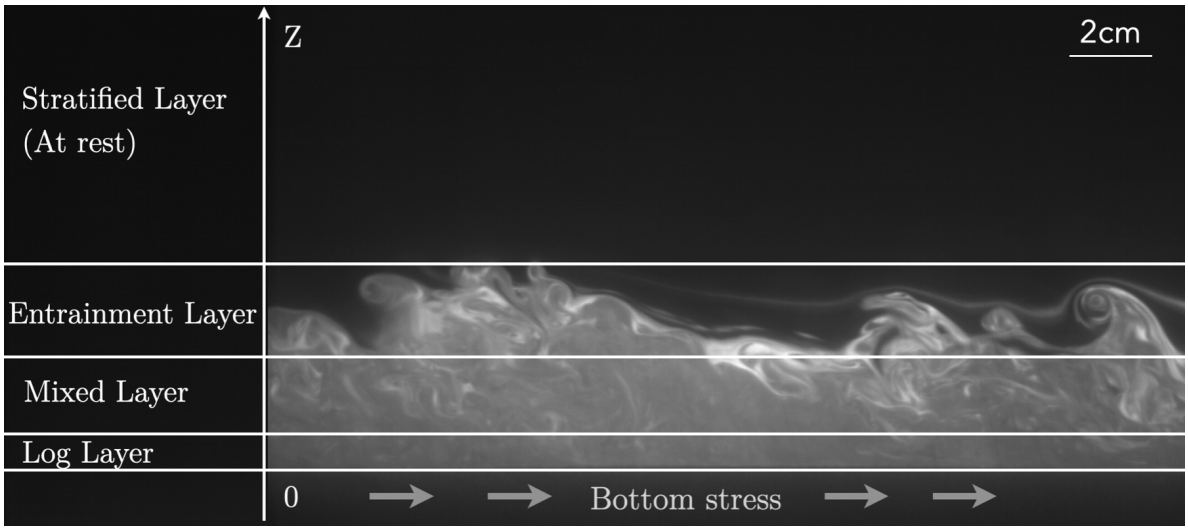


Figure 2. EXP-S: Snapshot of vertical view, at $r = 4.5$ m, of the boundary layer turbulence revealed by a fluorescent dye (in white) injected into the denser water at the bottom prior to the experiment. The horizontal lines qualitatively illustrate the vertical structure of the flow. The black region above is at rest. The gray arrows represent the direction of motion of the platform relative to the fluid reference frame.

by a continuous 5 W YAG laser source operating at $\lambda = 532$ nm, shaped into a vertical light sheet by a Powell lens. Two cameras (PCO Edge 5.5, 14-bit monochrome) are mounted on both sides of the light sheet, observing the same plane from two different oblique view angles (about 35 degrees with respect to the horizontal). Each camera provides the particle displacement in a plane perpendicular to its line of sight, so the three velocity components can be deduced by geometric projection. The method relies on a geometric calibration of the two view fields, finely tuned by the intercorrelation of the images transformed to physical coordinates.

Images are acquired in short bursts of 3 frames at 100 Hz, repeated five times per second. Processing is performed with the software UVMAT. Details on the S-PIV method are given in Sous et al. (2013). The pixel size corresponds to about 0.1 mm, and the spatial resolution of the PIV is about 10 pixels. This spatial resolution of 1 mm is also set by the laser sheet thickness, about 4 mm (projected to 2 mm on the images with the oblique field of view). The precision on displacement measured from the image correlation at each grid point is about 0.2 pixel, corresponding to 0.2 $\mathrm{cm\,s^{-1}}$. This adds a contribution to the measured TKE and Reynolds stress components of about 0.1 $\mathrm{cm^2\,s^{-2}}$. A few false velocity vectors also result from holes in the particle seeding. Most of them are eliminated by criteria of continuity and of consistency in the combination of the two camera views. The measurements of fluctuations are uncertain within 3 millimetres of the boundary because of strong shear, small spatial scale of turbulence and spurious illumination by the laser reflection on the wall.

The view field spans $L_{\mathrm{laser}} = 32$ cm along the azimuthal direction. The cartesian velocity components $(u_y, -u_x, u_z)$ in this tangential vertical plane can be treated as the cylindrical components (u_r, u_θ, u_z) to an excellent approximation. For a deviation angle θ with respect to the tangent point, the exact relation is

$$u_r = u_x \sin\theta + u_y \cos\theta \simeq \theta u_x + u_y \quad (3a)$$

$$u_\theta = -u_x \cos\theta + u_y \sin\theta \simeq -u_x + \theta u_y \,. \quad (3b)$$

At the edges of the view field, the maximum angular deviation is $\theta_{max} = \pm\, 0.035$ radian. The azimuthal velocity is therefore contaminated by 3.5 % of u_y and the radial velocity by 3.5 % of u_x. These small perturbations have opposite signs of both sides, so they cancel each other in the statistics over the whole width.

The velocity is measured in the platform reference frame which holds the cameras, but we systematically display velocities in the initial reference frame (where the outer bulk fluid is initially at rest). For that purpose, we subtract the velocity increment $v_0 = r_0\Delta\Omega = 23.6$ cm/s from the measured azimuthal component.

2.2.3. *Temperature measurements*

Temperature profiles are obtained using three vertically profiling thermistor probes mounted on a motor-driven gantry positioned along a tank radius (see Fig. 1). The probes are positioned at radial distances of 1.5 m, 2.5 m, and 4 m from the outer boundary. They are deliberately placed at the opposite azimuthal position with respect to the S-PIV measurement plane to minimise disturbance to the velocity field. Each probe traverses the water column at 5 cm s^{-1} with an acquisition frequency of 140 Hz. The raw profiles are separated by 10 s at a given depth and then interpolated onto a regular space-time grid.

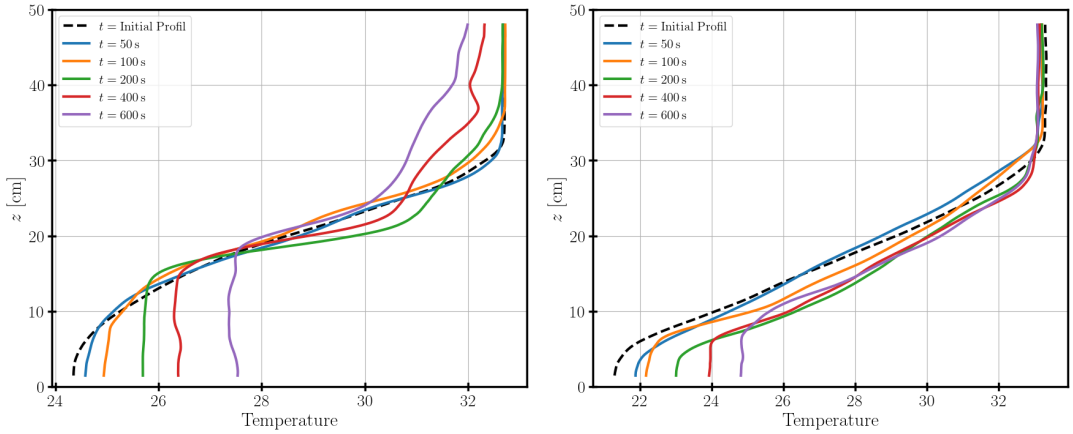


Figure 3. Vertical profiles of temperature at different time during the acquisition. Each color mark a different time of the experiment. The dark dashed line shows the initial profile of stratification prior to the experiment. For both figures, the temperature probe is at 4 m from the center of the tank. Left: Experiment without initial rotation (EXP-S). Right: Experiment with initial rotation (EXP-SR).

Figure 3 shows temperature profiles at several times during EXP-S (no rotation) and EXP-SR (with rotation). In both cases, the initial profile (dashed) exhibits a nearly linear temperature gradient over the lower 30 cm, capped by a well-mixed upper layer resulting from convection at the water–air interface. As the experiment progresses, the bottom stress erodes the stratification from below: the temperature profiles develop a growing isothermal layer near the floor, whose upper boundary marks the base of the mixed layer. The deepening is markedly faster in the non-rotating case (EXP-S), where the mixed layer reaches approximately 20 cm after 600 s, than in the rotating case (EXP-SR), where the Coriolis force limits the vertical penetration of momentum and the mixed layer remains significantly shallower over the same duration. This difference between the two experiments illustrates the controlling role of rotation on mixed-layer deepening and motivates the quantitative analysis presented in Section 4.

2.3. *Non-dimensional parameters*

We consider the flow produced in a cylindrical tank of radius $R = 6.5$ m and depth $H = 0.5$ m by a sudden change of rotation rate from Ω_i to $\Omega_f = \Omega_i + \Delta\Omega$. It is natural to express all horizontal length scales in terms of the measurement radius $r_0 = 4.5$ m $= 0.69\,R$. The relevant physical parameters are the rotation rate increment $\Delta\Omega$, the Coriolis parameter $f = 2\Omega_i$, the kinematic viscosity of water $\nu = 10^{-6}$ m^2 s^{-1}, the depth H, the radius r_0, and the time t,

the buoyancy frequency N, and the diffusivity of heat κ. We discuss here the relevant non-dimensional parameters which can be formed with these quantities.

The large aspect ratio, defined as,

$$a = \frac{r_0}{H} = 9(4)$$

ensures that the dynamics are controlled by local vertical fluxes, with negligible influence of the lateral boundary, at least during the initial phases of the spin-up. In the initial state of solid-body rotation, the free surface is in equilibrium with the centrifugal force and takes the shape of a paraboloid, with elevation $z = H + \Omega_i^2 r^2/(2g)$. For the present value $\Omega_i = 0.0524$ rad s^{-1}, the elevation difference between the center and the outer edge at $r = 6.5$ m is at most $\sim$5 mm, so the free surface can be treated as effectively flat.

We define the Rossby number Ro based on the rotation rate increment

$$Ro = \frac{\Delta\Omega}{\Omega_i}.(5)$$

In our experiments, $Ro \approx 1$ (the impulsive forcing is of the same order as the background rotation). We define the Reynolds number Re_δ based on the wall velocity $v_0 = r_0\Delta\Omega$ and the laminar Ekman layer thickness $\sqrt{\nu/\Omega_i}$:

$$Re_\delta = \frac{r_0\Delta\Omega}{\sqrt{\nu\Omega_i}}.(6)$$

This Reynolds number is $Re_\delta \approx 10^3$ in our experiments, well above the threshold for the transition to turbulence in Ekman layers ($Re_\delta \gtrsim 150$, Lingwood 1996, Sous et al. 2013), confirming that the boundary layer is fully turbulent. The same value $Re_\delta = 1000$ was used in the DNS of Marlatt et al. (2012), which provides evidence of a fully developed turbulent regime.

A more intrinsic Reynolds number can be formed from the friction velocity u_* and the turbulent Ekman layer thickness $\propto u_*/f$:

$$Re_* = \frac{u_*^2}{f\nu}.(7)$$

This quantity (sometimes called 'friction Rossby number'; e.g. Deardorff 1969) is not directly controlled by the experimental parameters but is estimated a posteriori from the measurement of u_* (see Eq. (1)). Its value $Re_* \approx 10^3$ lies well within the turbulent regime explored by Sous et al. (2013) and is comparable to values reported in DNS studies of the turbulent Ekman layer (Marlatt et al. 2012, Braun et al. 2020).

The Ekman number $E_k = \nu/(fH^2)$ also characterises the effect of viscosity in rotating flows. In the laminar regime it sets the Ekman layer thickness as $\delta/H = E_k^{1/2}$ and the Ekman spin-up time as $\Omega_i\tau_E = E_k^{-1/2}$. Its relevance in the turbulent regime is less clear. For our experiment, we have $Ek \approx \mathcal{O}(10^{-5})$. The Ekman number is related to other non dimensional parameters by $E_k = Ro^2 Re_\delta^{-2} a^{-2}$.

The time t is naturally expressed in units of the inertial period: $\Omega_i t$ when $\Omega_i \neq 0$, or $\Delta\Omega\, t$ otherwise. Three successive phases of the motion can be distinguished:

(i) $\Delta\Omega\, t < \pi$: the influence of fluid trajectory curvature is small and the flow can be treated as rectilinear.

(ii) $\Omega_i t \sim 2\pi$ (or $\Delta\Omega\, t \sim 2\pi$ if $\Omega_i = 0$): an Ekman layer (or von Kármán layer) is established after the damping of a few inertial oscillations, while the bulk flow remains largely unchanged.

(iii) $\Omega_i t \gg 2\pi$ (or $\Delta\Omega\, t \gg 2\pi$ if $\Omega_i = 0$): the bulk flow progressively spins up under the effect of bottom friction.

The values of the non-dimensional numbers can be compared to typical oceanic conditions. In the ocean bottom boundary layer over continental shelves, the friction velocity is of order $u_* \sim 0.5$–2 cm s^{-1} (Perlin et al. 2005) and the Coriolis parameter is $f \sim 10^{-4}$ s^{-1}. The turbulent Ekman depth $\delta_E \sim u_*/f$ is thus of order 10–100 m in the ocean, compared to $\sim$ 5–10 cm in our experiments. With the kinematic viscosity $\nu \sim 10^{-6}$ m^2 s^{-1} this yields $Re_* = u_*^2/(f\nu) \sim 10^7$–$10^8$ in the ocean, several orders of magnitude larger than in the laboratory. Even if the value of the Reynolds number is different in the ocean and in the experiment, our working hypothesis is that the experiments reproduce the main dynamical ingredients of a turbulent Ekman layer, including rotational confinement, stress-driven transport and inertial adjustment. Some Reynolds-number dependence may persist, particularly in the near-wall structure, but the experiments nevertheless provide a relevant dynamical analogue of oceanic boundary layers.

The influence of stable density stratification is quantified by the Richardson number based on the velocity $r_0\Delta\Omega$ and the fluid depth H:

$$Ri = \frac{N^2}{a^2\Delta\Omega^2}. \quad (8)$$

The Richardson number ranges from 0 (EXP-REF) to ≈ 0.4 (EXP-SR), leading to significant stratification effect at the scale of the whole fluid layer. We can also define a Richardson number Ri_* based on the Ekman layer thickness u_*/f instead of H,

$$Ri_* = \left(\frac{Nu_*}{fv_0}\right)^2. \quad (9)$$

In our experiments $N/f \sim 3$ while $u_*/v_0 \sim 1/20$, so that $Ri_* \sim 0.02$. Therefore, stratification has little influence in the dynamics of the Ekman layer. Temperature behaves as a passive scalar in this region.

The Prandtl number, ratio of kinematic viscosity to thermal diffusivity, is $Pr \simeq 6$ for water, in contrast with $Pr \simeq 700$ for salinity. Since the present study focuses on momentum dynamics and the stratification remains weak, the value of Pr has presumably a negligible influence on the results.

Alternative choices of non-dimensional parameters are possible. In particular, the final rotation rate Ω_f could be used instead of Ω_i. The general problem of a steady boundary layer between a fluid rotating at Ω_i and a plate rotating at Ω_f has been analysed by Lingwood (1996), providing both the steady solutions and their instability properties. This general case, called BEK (Bodewald-Ekman-Karman), encompasses as special cases the von Kármán layer ($\Omega_i = 0$), the Bödewald layer ($\Omega_f = 0$), and the Ekman layer recovered in the limit $\Omega_f \to \Omega_i$. Their analysis suggests using a weighted average of Ω_i and Ω_f, which reduces to $0.5(\Omega_i + \Omega_f)$ near the Ekman limit. However, the initial stage of spin-up studied here is more strongly influenced by Ω_i, which justifies our choice in Eqs. (5)–(6). This is further supported by Sous et al. (2013), who found experimentally that wall friction depends primarily on Ω_i in the turbulent regime.

3. Theory of the transient Ekman layer in the rotating tank configuration

3.1. *Extended 1D model for axisymmetric cases*

We can describe our experiments in terms of a one dimensional model along the vertical, starting from the Boussinesq equations for an axisymmetric flow in a rotating reference frame with Coriolis parameter f. In cylindrical coordinates, the incompressibility condition writes

$$\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{\partial u_z}{\partial z} = 0\,, \quad (10)$$

and the dynamical equations for the velocity components (u_r, u_θ, u_z) and the buoyancy $b = -g(\rho - \rho_0)/\rho_0$ write

$$\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + u_z\frac{\partial u_r}{\partial z} - \frac{u_\theta^2}{r} - fu_\theta + \frac{\partial p}{\partial r} = \frac{\partial}{\partial z}\left(\nu_t\frac{\partial u_r}{\partial z}\right) \quad (11a)$$

$$\frac{\partial u_\theta}{\partial t} + u_r\frac{\partial u_\theta}{\partial r} + u_z\frac{\partial u_\theta}{\partial z} + \left(f + \frac{u_\theta}{r}\right)u_r = \frac{\partial}{\partial z}\left(\nu_t\frac{\partial u_\theta}{\partial z}\right) \quad (11b)$$

$$\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + u_z\frac{\partial u_z}{\partial z} + \frac{\partial p}{\partial z} - b = \frac{\partial}{\partial z}\left(\nu_t\frac{\partial u_z}{\partial z}\right) \quad (11c)$$

$$\frac{\partial b}{\partial t} + u_r\frac{\partial b}{\partial r} + u_z\frac{\partial b}{\partial z} = \frac{\partial}{\partial z}\left(\kappa_t\frac{\partial b}{\partial z}\right), \quad (11d)$$

with p the dynamical pressure (pressure divided by a constant density ρ_0 in the Boussinesq approximation). Here, ν_t and κ_t denote the effective viscosity and diffusivity, respectively, accounting for both turbulent and molecular contributions. The horizontal turbulent fluxes are neglected owing to the large-aspect-ratio condition $a \gg 1$, while molecular transport is included implicitly in these effective coefficients to simplify the notation. In the rotating frame, the centrifugal force $\Omega^2 r$ is exactly balanced by a radial pressure gradient. The centrifugal force has therefore been omitted, and the pressure p in Eq. (11) is related to the total pressure p_{tot} by $p = p_{tot} - \Omega^2 r^2/2$.

Initially, the fluid and bottom plate ($z = 0$) are at rest (or in solid-body rotation). Taking this state as the reference frame, the velocity is zero everywhere for $t < 0$. At $t = 0$, the bottom plate is impulsively set to rotate at an angular velocity $\Delta\Omega$, while the velocity remains zero far from the boundary layer ($z \to \infty$). Since the geometry is axisymmetric, it is natural to seek solutions with u_r, u_θ proportional to r. Introducing a reference radius r_0, we write

$$u_r = \frac{r}{r_0}\,u(z,t) \;,\; u(0,t) = 0 \quad (12a)$$

$$u_\theta = \frac{r}{r_0}\,v(z,t) \;,\; v(0,t) = v_0 = r_0\,\Delta\Omega \;\text{ for } t > 0 \quad (12b)$$

$$u_z = w(z,t) \;,\; w(0,t) = 0 \quad (12c)$$

$$b = b(z,t) \;,\; \frac{\partial b}{\partial z}(0,t) = 0 \quad (12d)$$

Then Eq. (10) and (11) become

$$2\frac{u}{r_0} + \frac{\partial w}{\partial z} = 0 \quad (13a)$$

$$\frac{\partial u}{\partial t} + w\frac{\partial u}{\partial z} + \frac{u^2 - v^2}{r_0} - fv + \frac{r_0}{r}\frac{\partial p}{\partial r} = \frac{\partial}{\partial z}\left(\nu_t\frac{\partial u}{\partial z}\right) \quad (13b)$$

$$\frac{\partial v}{\partial t} + w\frac{\partial v}{\partial z} + \left(f + \frac{2v}{r_0}\right)u = \frac{\partial}{\partial z}\left(\nu_t\frac{\partial v}{\partial z}\right) \quad (13c)$$

$$p = p_\infty(r) + \int_z^\infty \left(\frac{\partial w}{\partial t} + w\frac{\partial w}{\partial z} - b\right)\,dz + \frac{w_\infty^2}{2} - \nu_t\left.\frac{\partial w}{\partial z}\right|_{z=0} \quad (13d)$$

$$\frac{\partial b}{\partial t} + w\frac{\partial b}{\partial z} = \frac{\partial}{\partial z}\left(\kappa_t \frac{\partial b}{\partial z}\right) \quad (13e)$$

For a viscosity ν_t and diffusivity κ_t depending only on z, this is a one-dimensional system of equations in terms of the coordinate z. The pressure is the sum of a term depending only on z (the hydrostatic pressure), and a term $p_\infty(r)$, equal to zero with our hypothesis of vanishing azimuthal velocity away from the boundary layer. The total pressure also includes the additional static contribution $\Omega^2 r^2/2$. It has no effect on the dynamics inside the fluid since its radial gradient exactly balances the centrifugal force, as stated above.

In summary this columnar model represents an Ekman boundary layer in unsteady regime, modified by the radial acceleration and transport effects due to the fluid rotation rate. The buoyancy has no direct action on the mean velocity. It acts only on the turbulent kinetic energy, which influences the mean velocity through the eddy viscosity.

Within the eddy viscosity hypothesis, the model is exact for a deep fluid layer in solid body rotation. It provides a radial mass flux that must return in the bulk because of the confinement by the lateral boundary. This drives a progressive spin-up by the associated Coriolis force. With the simple hypothesis of a vertically uniform radial velocity in the bulk, mostly valid in the absence of stratification, this yields a spin-up rate, as it will be further discussed in section 3.4. This effect is quadratic in velocity, so the bulk gradually deviates from the solid body rotation. Horizontal shear then develops. The model may still be appropriate with the local rotation rate, but we did not investigate those more general regimes.

Three phases of motion can be distinguished as stated in section 2.3.

(i) $\Delta\Omega\, t < \pi$: at short times, smaller than the half inertial period π/f, the radial velocity and non-linear terms in Eq. (13c) are negligible. The azimuthal velocity is driven by the bottom friction like in a unidirectional flow. Then the momentum balance can be expressed in terms of the displacement thickness

$$\delta_1 = \frac{1}{v_0}\int_0^\infty v\; dz \quad (14)$$

This thickness increases under the effect of wall stress u_*^2 as $d\delta_1/dt = u_*^2/v_0$. For a constant friction coefficient $C_\delta = u_*^2/v_0^2$, this results in the linear growth with time t,

$$\delta_1 = C_\delta\, v_0\, t \quad (15)$$

(ii) $\Omega_i t \sim 2\pi$ (or $\Delta\Omega\, t \sim 2\pi$ if $\Omega_i = 0$): After a time approaching half an inertial period, a radial velocity u appears, and the momentum balance is modified by the Coriolis term fu. This leads to inertial oscillations which relax to a steady Ekman layer, or to a more general Bodewald-Ekman-Karman layer if we include the local Coriolis effect associated with the angular velocity $\sim v_0/r_0$. This will be further discussed in section 5.4.

(iii) $\Omega_i t \gg 2\pi$ (or $\Delta\Omega\, t \gg 2\pi$ if $\Omega_i = 0$): At still later times the friction effect at the bottom is progressively transferred to the fluid column by Ekman pumping effects.

To close this 1D set of equations, turbulent viscosity and diffusivity will be computed with the k-ε model, implemented in the GOTM single-column framework (Umlauf and Burchard 2005). Full details of the numerical setup are given in Appendix A. Note that we do not solve Eq. (13) in its full form. First we neglect the radial pressure term, which is valid at the early stage, before the buildup of a significant bulk flow by the spin-up process. We also neglect the vertical advection of momentum which turns out to be a small correction, especially in the rotating case. The advection of TKE by the mean vertical velocity is also neglected.

3.2. *The transient Ekman layer in the laminar case*

To build physical intuition, we solve analytically the onset of the Ekman layer in the laminar limit ($Ro \to 0$, $Re_\delta \to 0$, $\nu = const.$). Two boundary conditions are considered: an imposed stress, whose exact solution has been already published by Ekman (1905), and an imposed velocity, which is the relevant condition for our laboratory tank and whose solution is derived here to complement Ekman's solution. The comparison of the two reveals that the boundary condition controls the damping rate of inertial oscillations. In the aforementioned limit, the system (13) reduces to

$$\frac{\partial u}{\partial t} - fv = \nu \frac{\partial^2 u}{\partial z^2}\,, \quad (16a)$$

$$\frac{\partial v}{\partial t} + fu = \nu \frac{\partial^2 v}{\partial z^2}\,. \quad (16b)$$

These two equations can be combined into a single equation by introducing the complex velocity $q = u + iv$, which satisfies

$$\frac{\partial q}{\partial t} + ifq = \nu \frac{\partial^2 q}{\partial z^2}. \quad (17)$$

Let us introduce the normalized coordinate z/δ_E, where $\delta_E = \sqrt{2\nu/f}$ is the thickness of the steady Ekman layer. Looking for time oscillating solutions in the form $q = \hat{q}(z)\exp(-i\omega t)$, we get spatial oscillations decaying with z of the form

$$\hat{q}(z) \propto \exp\left[-\frac{(1+i)}{(1-\omega/f)^{1/2}}\frac{z}{\delta_E}\right] \quad (18)$$

Note that the fluid is assumed at rest for large z since the flow is analysed in the rotating reference frame before spin-up, so that u, v exponentially decay for $z \to \infty$. In the alternative case of a uniform geostrophic velocity at large z, the same equation (16) would hold for the velocity defect, obtained by subtracting the geostrophic velocity to the velocity field.

The general solution of (17) can be expressed as an integral over frequencies of such elementary solutions. The weight of each frequency can be obtained as the Fourier transform of the step function representing the boundary condition at $z = 0$. After typically half an inertial period, this integral is quickly dominated by the frequency $\omega = 0$, dominant in the boundary condition, and by the inertial frequency $\omega = f$ which is resonant, as the penetration depth $\delta(\omega) = \delta_E(1-\omega/f)^{-1/2}$ diverges for $\omega \to f$. Then the solution can be approximately expressed as the sum of the classical steady Ekman layer solution $q_{\rm Ekman}(z)$ and an inertial oscillation whose amplitude $q_{\rm IO}(z,t)$ slowly evolves with time,

$$q(z,t) \simeq q_{\rm Ekman}(z) + q_{\rm IO}(z,t)\ \exp(-ift)\,. \quad (19)$$

The quality of this approximation will be checked below by the exact solution (Fig. 4) and compared with the laboratory experiments in Section 5.

The first term in Eq. (19) is the classical steady state Ekman solution given by (18) for $\omega = 0$, $q_{\rm Ekman}(z) \propto \exp(-z/\delta_E)\ \exp(-iz/\delta_E)$. Equation (18) diverges for $\omega = f$ but the inertial oscillation can be specified by introducing the form $q = q_{\rm IO}(z,t)\exp(-ift)$ in Eq. (17). This eliminates the Coriolis term and yields the pure diffusion equation

$$\frac{\partial q_{\rm IO}}{\partial t} = \nu \frac{\partial^2 q_{\rm IO}}{\partial z^2}. \quad (20)$$

It describes circular particle motion (in the anticyclonic direction) driven by the divergence of the diffusive flux. The amplitude of this circular motion diffuses with viscosity like the linear momentum in the absence of Coriolis effect. This contrasts with the steady component whose

penetration is inhibited by Coriolis effects. Therefore the inertial oscillation will eventually penetrate deeper than the steady component, forming a slab of circular motion whose thickness spreads as $\delta_{IO} \propto (\nu t)^{1/2}$.

The behaviour then depends on the boundary condition at $z = 0$. We shall distinguish the case of an imposed velocity v_0 and the case of an imposed shear stress u_*^2, for $t > 0$,

$$\begin{aligned} \text{imposed stress}: \quad & \nu \frac{\partial v}{\partial z} = -u_*^2 \,, \quad \nu \frac{\partial u}{\partial z} = 0 \\ \text{imposed velocity}: \quad & v = v_0 \,, \quad u = 0 \,. \end{aligned} \tag{21}$$

The steady component for a given velocity $v = v_0$ is given by

$$q_{\rm Ekman}(z) = i v_0 \, \exp\left[-(1+i)\frac{z}{\delta_E}\right] . \tag{22}$$

The corresponding viscous stress $-\nu \partial q_{\rm Ekman}/\partial z$ has the same form but is tilted by an angle $\pi/4$.

$$-\frac{\partial q_{\rm Ekman}}{\partial z} = i \frac{v_0}{\delta_E} \sqrt{2} \, \exp\left(i\frac{\pi}{4}\right) \, \exp\left[-(1+i)\frac{z}{\delta_E}\right] . \tag{23}$$

The value of this shear stress at the boundary is $u_*^2 = \nu v_0/(\delta_E \sqrt{2})$, tilted by $\pi/4$ with respect to the y axis. Therefore, Eq. (22) is also the solution of the problem with this imposed shear stress.

However, the inertial oscillation behaves differently in each case. Indeed, by vertical integration of (20), we get

$$\frac{d}{dt} \int_0^\infty q_{\rm IO} dz = -\nu \left. \frac{\partial q_{IO}}{\partial z} \right|_{z=0} . \tag{24}$$

A boundary condition of given shear stress $-\nu \left.\frac{\partial q}{\partial z}\right|_{z=0}$ will result in a zero average (over one inertial period) of the oscillating component $\nu \partial q_{IO}/\partial z = \nu \frac{\partial q}{\partial z} \exp(ift)$. Therefore, the integral $\int_0^\infty q_{\rm IO} dz$ is conserved. Since the thickness of the inertial oscillation increases by diffusion as $\delta_{IO} \propto (\nu t)^{1/2}$, the amplitude of the oscillation also decreases in $\propto (\nu t)^{-1/2}$. This argument does not apply to the imposed velocity solution: the full solution is needed to analyze the decay of the inertial oscillations.

We turn now to the exact solution of (17) for the stepwise surface forcing. The solution for the imposed stress has been published already in the seminal article of Ekman (1905) (the mathematical solution is acknowledged as a contribution of I. Fredholm). The approach relies on the Green's function of (17), which describes the response to an impulsive forcing applied at time $t_i < t$ and takes the form

$$q(z,t) = \frac{q_0}{\sqrt{t'}} \exp(-ift') \, \exp\left(-\frac{z^2}{4\nu t'}\right) , \tag{25}$$

where $t' = t - t_i$ (this is the elementary solution of the diffusion equation (20) for the variable $q \, \exp(ift)$).

Then the solution for the sudden shear stress set at $t > 0$ is obtained as an integral of this elementary solution over all time origins t_i in the range $[0 \;\; t]$. It yields the following integral over the time delay, expressed in terms of the non-dimensional time $\tau' = ft'$,

$$q(z,t) = \frac{i u_*^2}{\sqrt{\pi}(\nu f)^{1/2}} \int_0^{ft} \frac{\exp(-i\tau')}{\tau'^{1/2}} \, \exp\left(-\frac{f z^2}{4\nu\tau'}\right) \, d\tau' . \tag{26}$$

The prefactor has been adjusted to fit the shear stress condition at $z = 0$ as justified below. After taking the real and imaginary parts, it corresponds to Eq. (10) of Ekman (1905). This

is the exact solution of the transient problem. It can be interpreted as a sum of inertial oscillations with amplitude and phase depending on ζ.

The shear stress $-\nu\partial q/\partial z$ is then expressed as

$$-\nu\frac{\partial q}{\partial z} = \frac{iu_*^2}{\sqrt{2\pi}}\int_0^{ft}\frac{\exp(-i\tau')}{\tau'^{3/2}}\,\frac{z}{\delta_E}\,\exp\left(-\frac{fz^2}{4\nu\tau'}\right)d\tau' \,. (27)$$

This expression should tend to the step function (Heaviside function) in the limit $z \to 0$. This limit is not obvious since the integrand diverges for $\tau' \to 0$. It can be obtained by making the change of variable $X = (z/\delta_E)\tau'^{-1/2}$, such that $d\tau' = -2(z/\delta_E)^2 X^{-3} dX$. Then (26) becomes, reversing the bounds of the integral obtained with the new variable,

$$-\nu\frac{\partial q}{\partial z} = iu_*^2\sqrt{\frac{2}{\pi}}\int_{(z/\delta_E)(ft)^{-1/2}}^{\infty}\exp\left(-\frac{X^2}{2} - i\frac{z^2}{\delta_E^2 X^2}\right)\,dX \,. (28)$$

The integral now converges for any z, and for $z \to 0$, it tends to $\int_0^\infty \exp(-X^2/2)\,dX = \sqrt{\pi/2}$. Therefore, the imaginary part of $-\nu\partial q/\partial z$ is equal to u_*^2 for $t > 0$, while its real part is equal to 0, which fits with the boundary condition (21) for imposed stress.

The case of imposed velocity can be deduced from Eq. (26) by noticing that if $q(z,t)$ is a solution of (17), then $\partial q/\partial z$ is also a solution, thanks to the commutation of the partial derivatives with respect to time and space. Therefore, we switch from the case of imposed stress to the case of imposed velocity by the transform

$$q \;\to\; -\frac{v_0\nu}{u_*^2}\frac{\partial q}{\partial z} \,. (29)$$

Thus we immediately deduce from (27) a solution for an imposed velocity v_0,

$$q(z,t) = \frac{iv_0}{\sqrt{2\pi}}\int_0^{ft}\frac{\exp(-i\tau')}{\tau'^{3/2}}\,\frac{z}{\delta_E}\,\exp\left(-\frac{fz^2}{4\nu\tau'}\right)d\tau' \,. (30)$$

The main difference with the original solution (Eq. 26) derived by Ekman is the exponent of the decay time ($\tau^{-1/2}$ for the free-slip solution and $\tau^{-3/2}$ for the no-slip solution), such that the damping rate is very different in the two cases.

We have numerically determined the solution (26) for an imposed wall stress and the solution (30) for the imposed velocity case, using the reduced vertical coordinate $\zeta = z/\delta_E$. For simplicity, we refer to the imposed wall-stress and imposed-velocity cases as the free-slip and no-slip cases, respectively. In the no-slip case, we choose a unit velocity $v_0 = 1$, and in the free-slip case a wall condition $-\partial|q|/\partial\zeta = \sqrt{2}$ tilted by an angle $\pi/4$ so that the final steady state is the same in both cases. We plot in Fig. 4 the two velocity components at altitude $\zeta = 1$ (altitude at which the two velocity components are equal in the steady regime). As anticipated, we observe a much faster decay of the inertial oscillations in the no-slip case. Although we have not made any hypothesis about the separation of the steady regime and inertial oscillation, the exact solution appears as a clear superposition of the two solutions. The same result holds at various heights ζ, even if the amplitude and phase of the inertial oscillations are different from the one plotted in Fig. 4 (not shown).

3.3. *The transient Ekman layer in the turbulent case*

A key question is how these laminar results extend to the turbulent regime. In the limit of large Reynolds numbers, steady flow components are expected to be functions of the reduced vertical coordinate $\zeta = zf/u_*$. This reflects the hypothesis that the properties of turbulence are fully determined by the friction velocity u_* (the eddy viscosity hypothesis is not required). This yields a universal velocity profile, the turbulent counterpart of (22), which must be matched with the log layer near the wall (see for instance Garratt 1992).

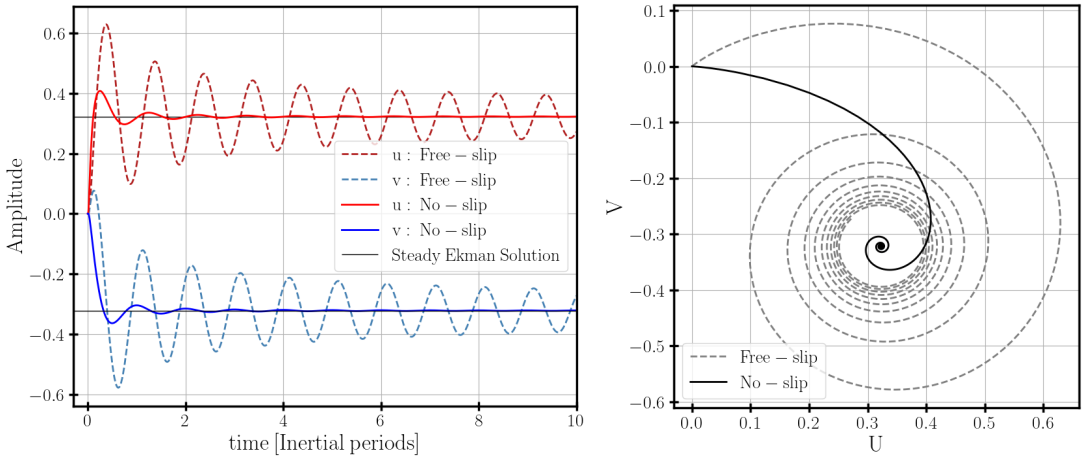


Figure 4. Time evolution of the horizontal velocity at $\zeta = 1$ for the no-slip and free-slip boundary conditions. Left: Solution of Eq. (30) (solid lines), corresponding to an imposed velocity in the x-direction, and solution of Eq. (26) (dashed lines), corresponding to an imposed shear stress oriented at an angle $\pi/4$. The thin black lines indicate the steady Ekman solution given by Eq. (22). Right: Hodographs of the horizontal velocity over ten successive inertial oscillations at $\zeta = 1$, for the no-slip (solid line) and free-slip (dashed line) boundary conditions.

Although the time-dependent eddy viscosity $\nu_t(z,t)$ couples the mean flow to the oscillating shear, the steady-state Ekman balance is independent of the transient excitation: provided any damping mechanism is present, the inertial oscillations decay and the flow converges to the same steady solution, regardless of the damping timescale. This will be investigated in Section 5 by decomposing the velocity field into a time-mean component and an oscillating residual, and comparing the k-ε simulations with the experimental results.

One crucial aspect of the numerical model is the representation of the eddy viscosity near the wall. In a turbulent boundary layer, the eddy viscosity ν_t vanishes as $\nu_t \sim \kappa u_* z$ in the log layer and drops to zero at $z = 0$, where the flow is locally laminar. The thickness of this laminar sublayer is classically estimated as $5\nu/u_*$, about 0.5 mm in our experiments. This is difficult to resolve numerically as it requires adaptative grid mesh and specific adaptation of the turbulent closure, as proposed for instance by Speziale et al. (1992). As an alternative, we choose a first grid point in the turbulent region, at an altitude $z_1 = 1$ mm, and impose a shear stress which follows a quadratic drag law

$$\tau_b = -C_f |\mathbf{u}(z_1) - \mathbf{u_{wall}}| \, (\mathbf{u}(z_1) - \mathbf{u_{wall}}), \quad (31)$$

This drag law encodes the integrated effect of the unresolved viscous sublayer through the empirical coefficient C_f whose determination is discussed in the appendix A .

The damping rate of the inertial oscillations is therefore controlled by C_f. In the free-slip limit, $C_f = 0$ and there is no damping from the boundary: the inertial oscillations only slowly decay as they extend in thickness, with conservation of their global momentum. In the no-slip case, the friction coefficient introduces additional damping. We will compare this parameterization with the experiments in sections 4 and 5.

3.4. *Ekman pumping effects*

The spin-up process in the bulk depends on the recirculation of the vertical flow produced in the boundary layer, which forms the Ekman pumping. In the absence of buoyancy effects, the resulting azimuthal velocity away from the wall $v_\infty(r,t)$ can be assumed independent of z. This results from the shallow water geometry, with height H smaller than the radius r_0, and the Taylor column effect due to rotation. Then the time evolution of the bulk azimuthal

velocity v_∞ can be obtained by vertical integration of Eq. (11b). In a first approximation the water column momentum $\int_0^H u_\theta dz \simeq H v_\infty$ is locally driven by wall friction

$$H\frac{\partial v_\infty}{\partial t} = -\nu_t \left.\frac{\partial u_\theta}{\partial z}\right|_{z=0} = u_*^2 \cos\alpha_0 \,. (32)$$

To refine this expression, we need to take into account additional terms due to advection by vertical and radial velocities. Assuming a fixed upper surface, the integral of u_r then vanishes by mass conservation. Furthermore, using (10), we can express $u_z \partial u_\theta/\partial z = \partial(u_\theta u_z)/\partial z + (u_\theta/r)\partial(r u_r)/\partial r$. The vertical integral of the first term vanishes by the boundary conditions $u_z = 0$ at $z = 0$ and $z = H$, while the second term adds to $u_r \partial u_\theta/\partial r$. Then, the vertical integral of Eq. (11b) can be expressed as

$$H\frac{\partial v_\infty}{\partial t} = u_*^2 \cos\alpha_0 - \frac{1}{r^2}\frac{\partial}{\partial r}\int r^2 u_r u_\theta \, dz (33)$$

The second term represents the divergence of the radial advection of angular momentum. With the scaling (12), the latter is equal to $-(4/r_0)\int uv dz$. Since $\int u dz = 0$, this can be also written as $-(4/r_0)\int u(v - v_\infty)dz$. For spin-up, the radial velocity within the boundary layer is outward ($u > 0$), while $v - v_\infty > 0$, so that it is opposed to the driving force[1]. This radial transport term is quadratic in velocity like the wall friction u_*^2. Therefore, the right hand term of (33) can be empirically modelled as

$$H\frac{dv_\infty}{dt} = C_b(v_0 - v_\infty)^2 \,, (34)$$

where v_0 is the wall velocity. This integrates in time as

$$v_\infty = v_0 \frac{(C_b v_0/H)\, t}{1 + (C_b v_0/H)\, t}, (35)$$

This introduces a characteristic turbulent spin-up time $\tau_{\rm turb} = H/(C_b v_0)$, which must be compared to the inertial time Ω_i^{-1} to assess the relative importance of rotation and friction in the long-time evolution. The coefficient C_b will be determined experimentally. In the stratified case, the Ekman pumping competes with the buoyancy effects, so the spin-up is limited to a layer smaller than H. We will not consider this case here.

4. Mean flow and turbulent statistics

The theoretical developments give us insights into the dynamical regimes that we can expect in the transient Ekman layer. We now examine the mean flow structure and turbulence statistics in the laboratory experiments and assess how well they are captured by the k-ε model.

4.1. *Coherent structures*

Figure 5 shows dye visualisations of the boundary layer with and without background rotation, offering a qualitative picture of how rotation reorganises the turbulent boundary layer. In the absence of rotation (Fig. 5, top), large vigorous structures erupt from the bottom and penetrate deep into the interior, reaching heights of 10–12 cm. The dye interface is highly irregular, with mushroom-shaped tilted plumes and overturning billows across a broad range of scales, consistent with the absence of a rotational constraint. Similar patterns are visible

[1]For spin-down, $u < 0$, while $v - v_\infty < 0$, so it enhances the driving force (negative in this case).

in DNS computations of the temporally developing turbulent boundary layer by Kozul et al. (2016).

In the rotating case (Fig. 5, bottom), turbulent activity is confined to a layer of 6–8 cm, consistently with the rotational scaling $\delta_E \sim u_*/f$. The structures are markedly smaller and more regular, the interface sharper and more horizontal, indicative of the suppression of large vertical excursions by rotation.

4.2. *Mean flow profiles*

We now turn to a more quantitative analysis of the boundary layer structure, focusing on three configurations: EXP-REF, EXP-S and EXP-SR (cf. Table 1). Stereoscopic PIV measurements provide the three velocity components in a vertical plane aligned with the azimuthal direction(see section 2.2.2 for details). The velocity field has been decomposed into a mean and fluctuating component, with respect to horizontal average across the view field (32 cm wide). The resulting mean velocity and Reynolds stress components are then averaged on a sliding time window to improve the statistics. The results can be interpreted as 1D profiles that allow the comparison with the 1D numerical simulation.

We plot the azimuthal velocity for the three configurations at $t = 100$ s in Fig. 6. The averaging time window has been chosen as 20 s in the absence of rotation and 60 s, an inertial period, in the rotating case (the results of a shorter time window 10 s are very similar but noisier). The shear is strongest near the wall, but the spatial resolution is insufficient to capture the wall velocity, $v_0 = 23.6$ cm/s at $z = 0$ (at rest with respect to the camera). In the non-rotating cases, the near-bottom velocities are similar in the two experiments. The boundary layer is slightly thinner in the stratified case, although the difference is small. For the rotating case, $t = 100$ s corresponds to a stage at which the inertial oscillations have been largely damped, while the spin-up of the bulk is still in its early stages (see Fig. 16). To account for the fact that the spin-up of the interior has already begun, we subtract the interior velocity, denoted by u_θ^{top}, from the velocity profile. We refer to the resulting quantity, $u_\theta - u_\theta^{top}$, as the velocity defect. In practice, the interior velocity u_θ^{top} is obtained by averaging the velocity over the uppermost centimetre of the available profile, at $z = 13$ cm.

The main differences between the non-rotating and rotating boundary layers appear in the layer above the near-bottom region. Rotation constrains the vertical extent of the boundary layer, with a super-geostrophic overshoot visible between 2 and 5 cm ($\langle u_\theta \rangle - u_\theta^{top} < 0$) , whereas the non-rotating boundary layer extends beyond the observation window. The height at which this super-geostrophic velocity reaches its maximum amplitude is defined as $\delta_{U\,\max}$.

This observation is also apparent in the radial velocity profiles (Fig. 7). In the rotating case, the radial velocity is driven by the deflection of the azimuthal circulation by the Coriolis force, and the shear is approximately constant throughout the layer, in contrast with the azimuthal component. A small radial velocity is also produced in the non-rotating case (EXP-Ref and EXP-S), under the centrifugal effect of the rotating plate. A slight confinement by stratification is observed with a negative part in the upper part. This presumably reflect the radial recirculating flow produced by the Ekman pumping, which is vertically confined by stratification.

The Ekman pumping also results in a small downward mean velocity, which can be estimated from the integral of the radial velocity by Eq. 13. This yields a value -0.05 $\mathrm{cm\,s^{-1}}$ in the rotating case and -0.07 $\mathrm{cm\,s^{-1}}$ in the non-rotating case. We measure a small negative velocity in the range -0.05 to -0.2 $\mathrm{cm\,s^{-1}}$. This implies an experimental error on the stereoscopic combination of the order of 0.15 $\mathrm{cm\,s^{-1}}$, less than 1 % of the maximum velocity.

These velocity profiles are well reproduced by the 1D k-ε model in the rotating case. The region of super-geostrophic overshoot ($\langle u_\theta \rangle - u_\theta^{top} < 0$) nevertheless extends further away than

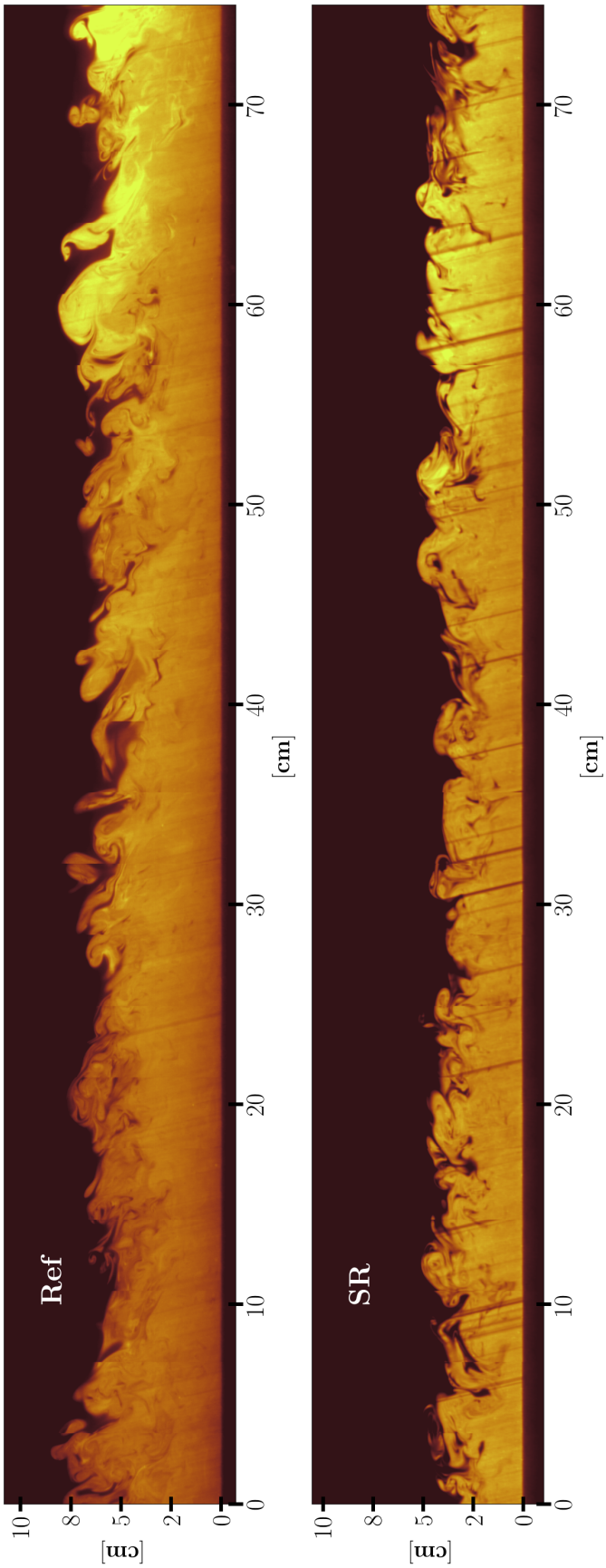


Figure 5. Panoramic reconstruction of the dye-visualised boundary layer over a horizontal extent of 0.75 m, obtained by compositing successive frames spanning $\Delta t = 5$ s. The first frame of each composite is taken at $t = 52$ s after the impulsive spin-up. Note that the contrast has been adjusted independently for each panel to enhance the visibility of the structures. Top: non-rotating, unstratified case (EXP-REF). Bottom: rotating, weakly stratified case (EXP-SR). For both picture, the saturation on the right comes from a brighter illumination of the right part of the laser sheet. Inclined black lines are shadows from clusters of particles on the surface.

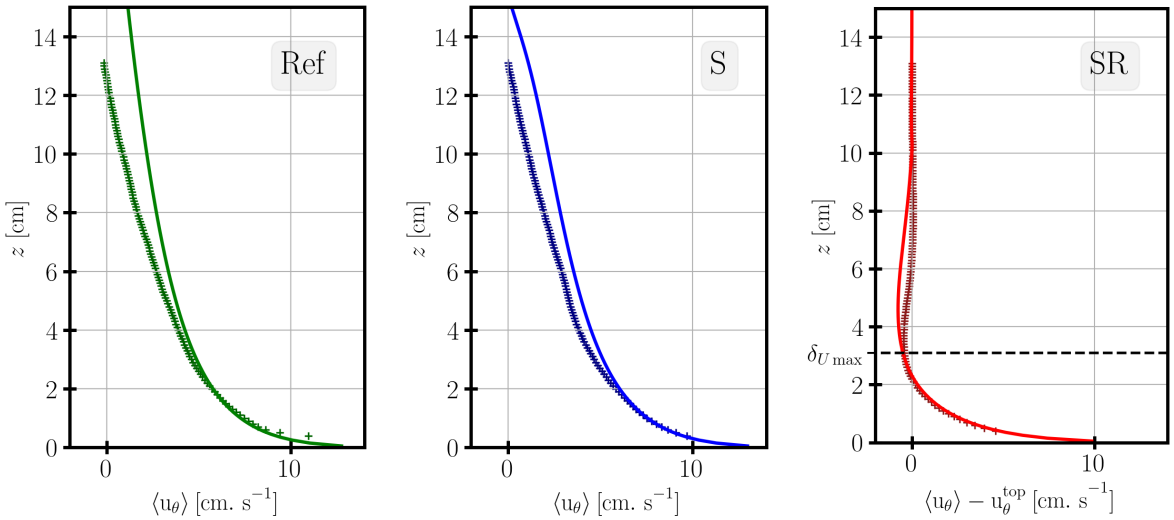


Figure 6. Vertical profiles of azimuthal mean velocity $\langle u_\theta \rangle$ at $t = 100$ s. Experimental data are marked by +. We plot with solid line, the vertical profile from k-ε simulation. The wall stress velocity is $\simeq$ 1 cm/s, so z=1 cm corresponds to $y^+ = z\nu/u_* \simeq 100$ in wall units. Left: Experiment without rotation-without stratification (EXP-REF). Center: Experiment without rotation-with stratification (EXP-S) . Right: Experiment with rotation-with stratification (EXP-SR). For the experiment with rotation, the interior velocity (u_θ^{top}) has been subtracted in order to visualize the velocity anomaly. The horizontal dashed line in the right panel indicates the height of $\delta_{U\,\max}$ where the overshoot is at its maximum value.

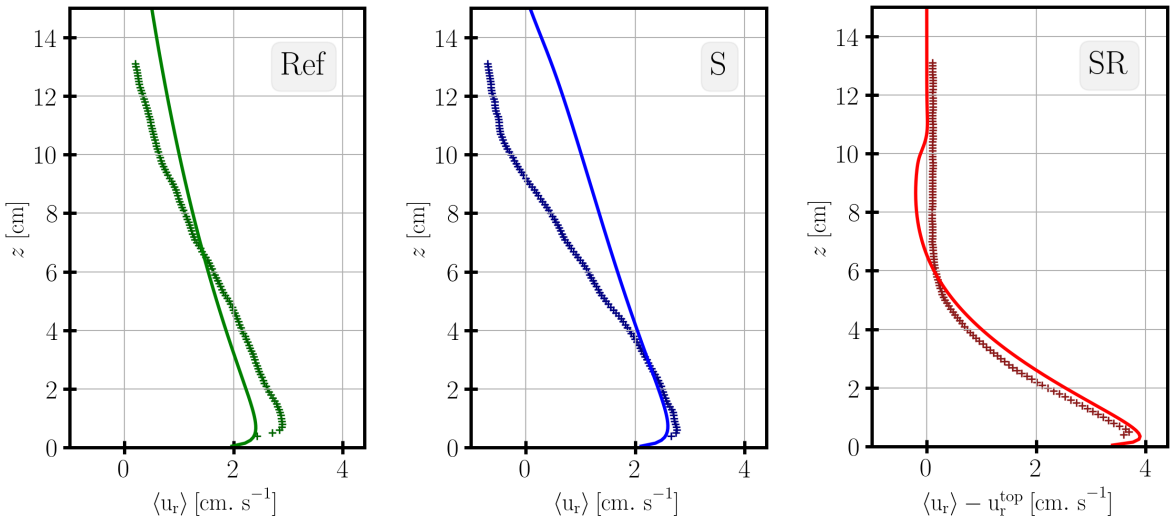


Figure 7. Same as figure 6 but for the radial velocity at $t = 100$ s

in the experiment, which may reflect the inability of the model to represent the intermittent nature of turbulence at the edge of the boundary layer. Note also that the global agreement depends on the choice of the friction coefficient C_f (see appendix A). A stronger friction would increase the near wall velocity while increasing the boundary layer thickness according to the scaling u_*/f.

We find also a good agreement of our mean velocity profiles with respect to the DNS of Marlatt et al. (2012) (see Fig. 8). This confirms that the turbulent boundary layer can be considered as an Ekman layer in the reference frame of the fluid, in spite of the large Rossby number $\Delta\Omega/\Omega_i = 1$.

In the non-rotating case, the agreement between the 1D k-ε model and the experiments is less accurate, which is presumably due to the neglection of the vertical advective term, which squeezes the boundary layer. Despite these limitations, the 1D model captures the amplitude

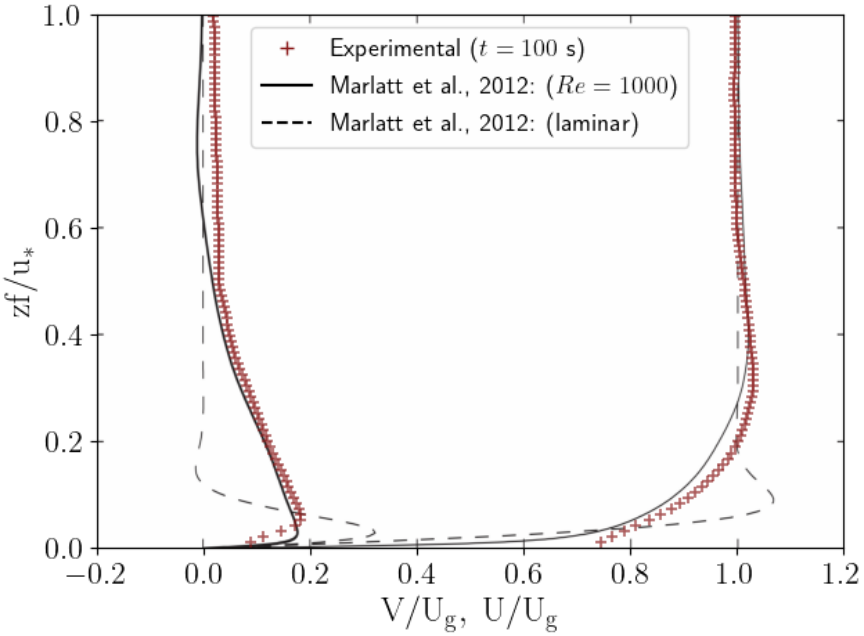


Figure 8. Comparison of the normalized mean velocity profiles with the Ekman-layer solutions of Marlatt et al. (2012)(see their Fig 3). Experimental profiles at $t = 100$ s are shown by crosses. The vertical coordinate is normalized using $u_* = 1$ cm s^{-1} and $f = 0.1$ s^{-1}, together with the turbulent DNS at $Re = 1000$ (solid lines) and the laminar Ekman solution (dashed lines).

and general shape of the vertical profile reasonably well. The similarity between the profiles in EXP-REF and EXP-S confirms that the chosen stratification does not exert a strong control on the dynamics.

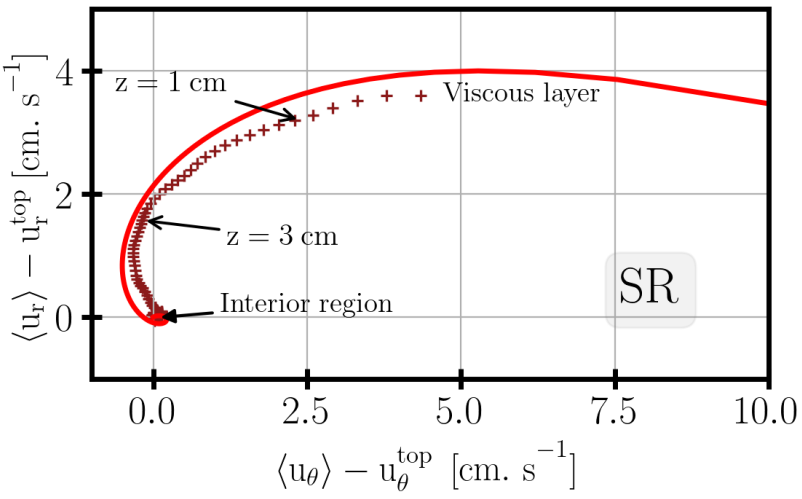


Figure 9. Hodograph of the defect velocity at $t = 100$ s showing the Ekman Spiral (radial vs azimuthal velocity components). Different height (in cm) are indicated by black arrow as the two characteristic region – the Interior region and the Viscous layer. In solid line, the vertical profile from k-ε simulation. Experimental data (SR) are plotted with crosses (+).

The hodographs corresponding to these velocity profiles are shown in Figure 9. The Ekman spiral is clearly visible over the observed portion of the water column. The hodograph does not extend to the bottom boundary, so the viscous sublayer and the cross-isobaric angle cannot be directly determined from this plot alone; the missing endpoint corresponds to the no-slip condition at the tank bottom, where $u_r = 0$ and $u_\theta = v_0$. The k-ε model reproduces the

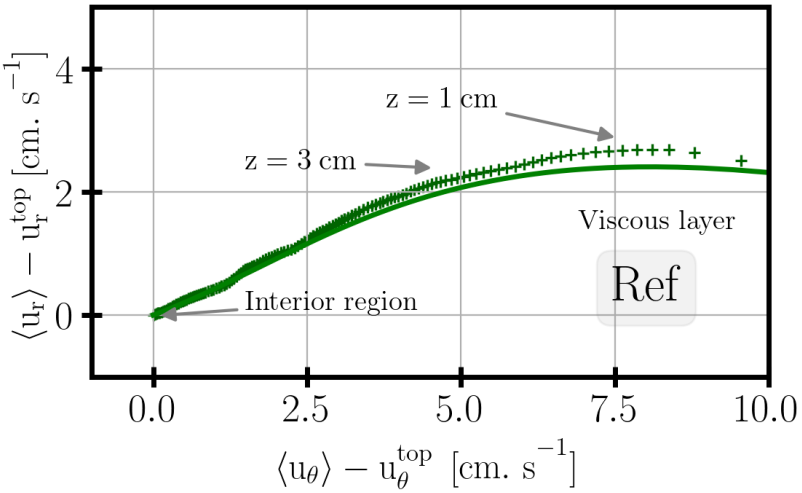


Figure 10. Hodograph of the defect velocity at $t = 100$ s showing the Ekman Spiral (radial vs azimuthal velocity components). Different height (in cm) are indicated by black arrow as the two characteristic region – the Interior region and the Viscous layer. In solid line, the vertical profile from k-ε simulation. Experimental data (EXP-REF) are plotted with crosses (+). For this figure only, we also subtract the top velocity.

overall spiral structure fairly accurately.

For comparison, the hodograph obtained for the non-rotating experiment EXP-REF is shown in Figure 10. Unlike the rotating case, no significant variation of the angle between the azimuthal and radial velocity components is observed across the boundary layer, particularly away from the wall.

The vertical confinement by rotation is also observed in the temperature profiles (Fig. 11). The mixed layer reaches only 6 cm in the rotating case against 13 cm in the non-rotating case despite similar initial stratification. The k-ε model reproduces well the overall behavior, although the thermocline interface is sharper than observed in the experiments. This effect that is particularly pronounced in the rotating case.

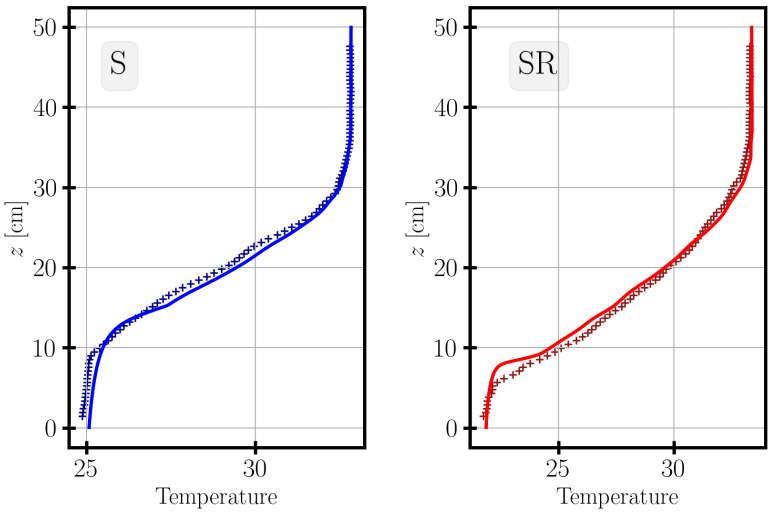


Figure 11. Same as figure 6 but for the temperature profile at $t = 100$ s. The vertical axis is greater than the one for the velocity data.

4.3. *Turbulent quantities*

Figure 12 shows the vertical profiles of TKE at $t = 100$ s for the three configurations. In all cases, the TKE decreases monotonically with height, with maximum values near the boundary of order 2 $\mathrm{cm}^2\,\mathrm{s}^{-2}$, reflecting the dominance of bottom-generated shear turbulence. Our first reliable data point is at $z = 0.4$ cm, so we do not capture the decay in the buffer layer. In the non-rotating unstratified case (EXP-REF, left panel), the TKE remains significant up to the top of the observation window, consistently with the absence of mechanism limiting the vertical extent of turbulence. In the rotating stratified case (EXP-SR, right panel), the TKE drops more sharply above $z \approx 6$ cm, consistently with the rotational confinement of the boundary layer.

The k-ε model reproduces well these profiles, considering the limited precision of these measurements. On one hand, the experimental TKE is underestimated near the wall by the spatial filtering of the smallest scales. This is due in particular to the 4 mm thickness of the laser sheet which is not small in comparison with the integral scale of the turbulence, of the order of 1 cm. On the other hand the TKE is increased by the measurement noise, of order 0.1 $\mathrm{cm}^2\,\mathrm{s}^{-2}$, which leads to an overestimation at large altitude. In the absence of rotation, the boundary layer thickness is definitely overestimated by the model, as it was observed for the mean velocity, probably because of the absence of vertical advection.

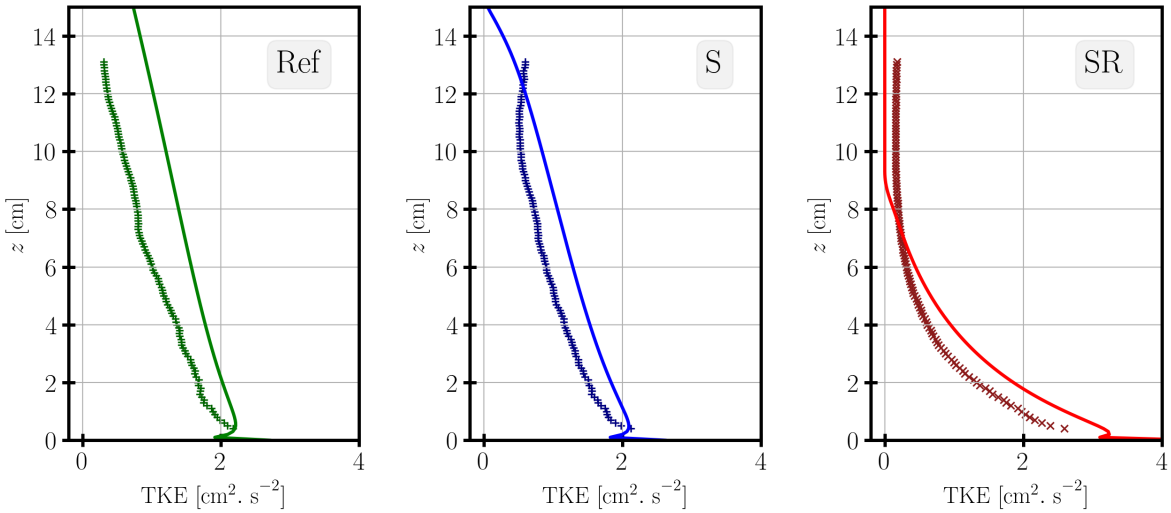


Figure 12. Vertical profiles of TKE at $t = 100$ s. Experimental data, marked by $\times$ are averaged, for each height, over a 20 s temporal window for REF and S cases and 60 s for the rotating case SR. In solid line, the vertical profile from k-ε simulation. Left: Experiment without rotation-without stratification (case REF). Center: Experiment without rotation-with stratification (case S). Right: Experiment with rotation-with stratification (case SR).

The variance of the three velocity components that compose the TKE is plotted in Fig. 13. In the strongly sheared zone near the wall, the variance of the azimuthal component dominates over the vertical component typically by a factor of 3, while the radial component lies in between. The turbulence becomes more isotropic away from the wall. These results are consistent with what is known in usual turbulent shear layers. Note that the k-ε model deals only with TKE, without distinguishing between the velocity components as dynamical variables.

Figure 14 shows the vertical profiles of the azimuthal turbulent momentum flux $\langle u'_\theta u'_z \rangle$ for the three configurations at $t = 100$ s. In the non-rotating cases (left and middle panels), the flux increases from zero at the bottom boundary, reaches a broad maximum of approximately 0.6 $\mathrm{cm}^2\,\mathrm{s}^{-2}$ near the bottom of the observation window, and decays linearly throughout the mixed layer. We use the extrapolation of this linear profile to get the value of the azimuthal shear stress $u_*^2 \cos\alpha_0 \simeq 0.6$ $\mathrm{cm}\,\mathrm{s}^{-1}$. The k-ε model roughly reproduces this profile, although with a larger shear stress and layer thickness.

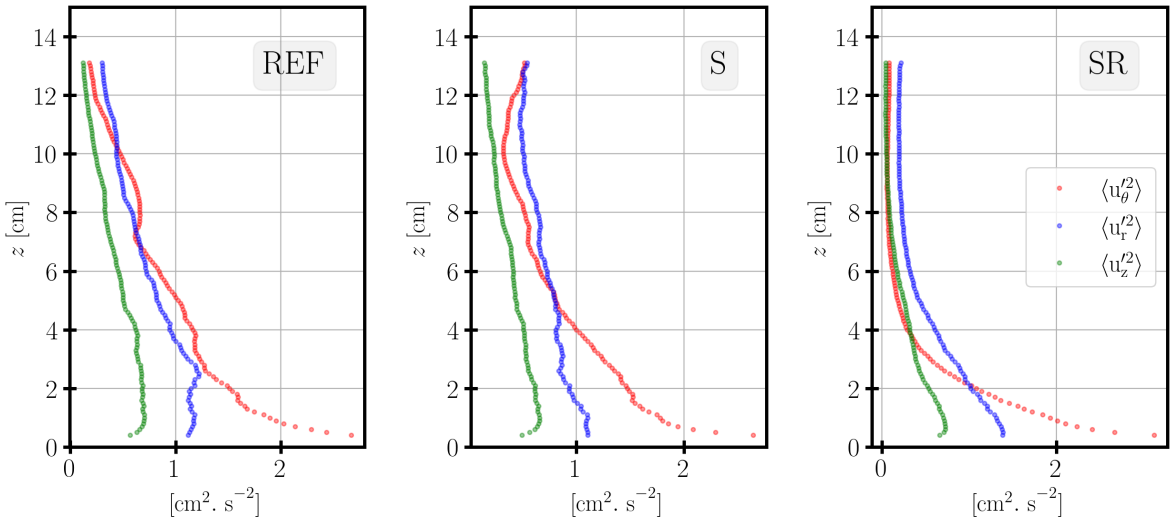


Figure 13. Vertical profiles of the three components of the TKE measured in the experiments.

In the rotating stratified case (right panel), the turbulent flux is confined below $z \approx 6$ cm, consistently with the rotational suppression of the boundary layer depth observed in the mean velocity profiles (Fig. 6) and the mixed-layer thickness (Fig. 11). The flux peaks close to the bottom boundary and decays rapidly with height, reflecting the concentration of shear production within the Ekman layer. Above the layer, the flux drops to near zero, indicating that turbulent momentum transport is effectively shut off by the combined effect of rotation and stratification. The k-ε model also captures this confinement fairly accurately. The experimental results remain significantly lower than the model near the wall, which is partially due to the spatial filtering of the measurements, as stated above. Moreover, the model does not include the viscous contribution to the wall stress (as specified in Appendix A), so an excess of eddy flux is needed to provide a good overall agreement. The viscous contribution can be estimated from the velocity shift $\langle u_\theta \rangle - v_0 \simeq 18$ cm s^{-1} at z=0.4 cm, which yields a gradient 42 s^{-1}, hence a viscous force $\nu \partial u/\partial z \simeq 0.4$ cm s^{-1}, so the total wall friction is $u_*^2 \cos\alpha_0 \simeq 1.2$ cm^{2}s^{-2}. This fits with the boundary value obtained in the model (see also section 5.1).

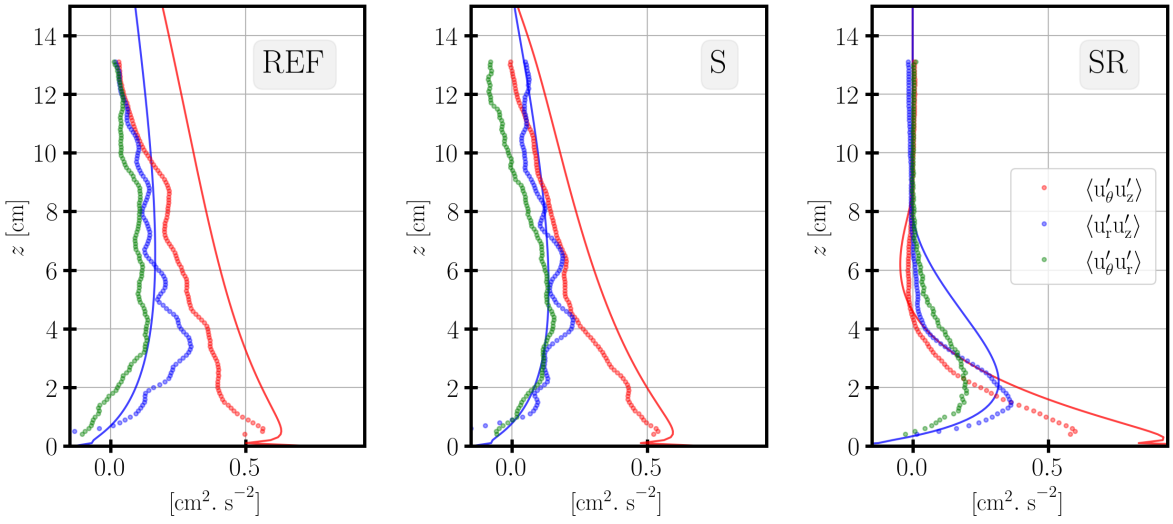


Figure 14. Vertical profile of the Reynolds stress component $\langle u'_r u'_z \rangle$, $\langle u'_\theta u'_z \rangle$ with dots compared to the k-ε model in solid lines, and correlation $\langle u'_r u'_\theta \rangle$.

The radial correlation components $\langle u_r u_z \rangle$ is also in rough agreement with the k-ε model, as

shown in Figure 14, although the maximum is confined to a smaller altitude. This figure also shows the correlation $\langle u_r u_\theta \rangle$, which is of the same order as $\langle u_r u_z \rangle$. The latter does not appear in a 1D vertical model. It reveals an outward turbulent flux of angular momentum, which is however small compared to the outward flux associated to the mean radial flow. Note also that these results are in fair agreement with the DNS of Marlatt et al. (2012) for the turbulent Ekman layer (the signs of the correlations are deduced by mirror symmetry $u_\theta \to -u_\theta$ and u_r, u_z unchanged).

We can deduce eddy viscosities along the azimuthal and radial directions from these velocity correlations, by dividing them by the local vertical shear. This gives the quantities $\nu_\theta = -\langle u_z u_\theta \rangle / (d\langle u_\theta \rangle / dz)$ and $\nu_r = -\langle u_z u_r \rangle / (d\langle u_r \rangle / dz)$ respectively. The k-ε model assumes that these two eddy viscosity are equal, so this is a good direct test of its validity. The comparison is shown in Figure 15. In the absence of rotation, the radial component is noisy because of the weak shear, but it fluctuates around the profile of the azimuthal component. The results are more reliable in the presence of rotation. Then the azimuthal component diverges at $z \simeq 3.5$ cm, corresponding to the velocity maximum, with vanishing vertical derivative. We can observe in Figure 14 that the Reynolds stress indeed changes sign in the upper region of the boundary layer, in relation with the reversed shear, but at a slightly higher level as the velocity maximum. Below this divergence point, the radial eddy viscosity remains slightly lower than the azimuthal component but this may be within the experimental errors. We can conclude that the isotropic eddy viscosity assumption of the model is good in the most active zone. The eddy viscosity of the model seems to be in excess in the outer part of the boundary layer, as mentioned earlier. However this has little influence in the prediction of the flow evolution.

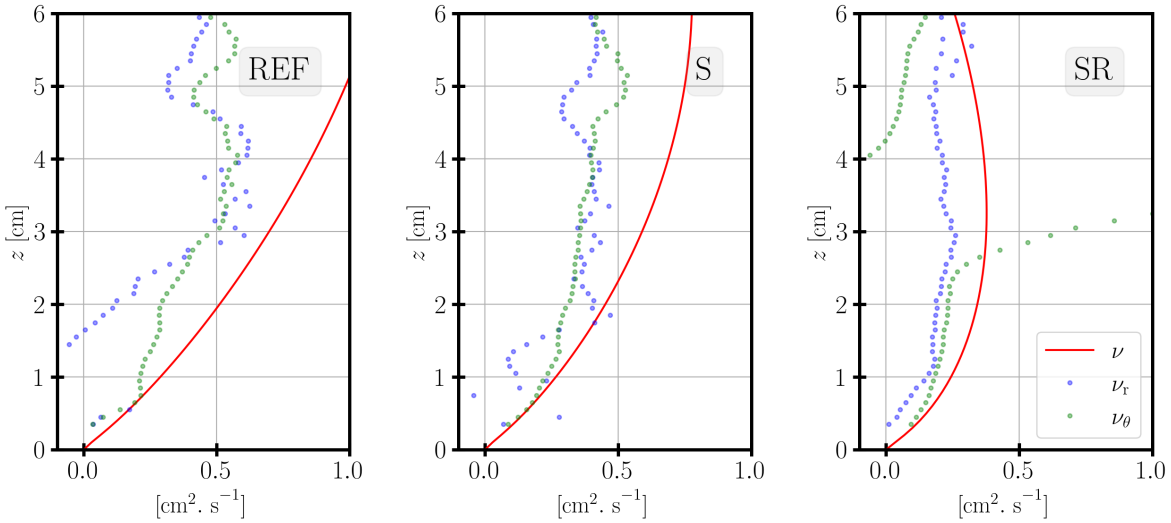


Figure 15. Vertical profiles of the measured eddy viscosities $\nu_\theta = -\langle u_z u_\theta \rangle / (d\langle u_\theta \rangle / dz)$ and $\nu_r = -\langle u_z u_r \rangle / (d\langle u_r \rangle / dz)$ with dots compared to the results of the k-ε model in solid line.

The turbulent statistics presented above reflect the same physical picture as the mean profiles: in the non-rotating cases, turbulence fills the entire mixed layer and drives efficient momentum transport, while in the rotating case it is confined within the Ekman layer. This contrast between scalar homogenisation and persistent velocity gradients is physically meaningful: scalars are mixed down-gradient by turbulent diffusion alone, whereas momentum is subject to the additional constraints of rotation and the time-evolving bottom stress. The following section examines how these dynamics evolve in time.

5. Transient evolution of the turbulent Ekman Layer

5.1. *Bulk Spin-up adjustment*

Figure 16 shows a time series of the azimuthal velocity, averaged over the uppermost centimetre of the vertical profile u_θ^{top}. This quantity serves as a proxy for the interior geostrophic velocity noted v_∞ in section 3.4. In the fluid frame, it increases monotonically toward the platform velocity v_0 as bottom friction spins up the interior flow.

The dashed line shows the friction law (Eq. 35), which provides a good fit to the observed spin-up, yielding $C_b \simeq 2.5 \times 10^{-3}$. The difference between C_f and C_b is expected: C_f is defined with respect to the near-wall velocity at z_1, which is smaller than the depth-averaged velocity used in the bulk formulation, so that $C_f > C_b$. Low-frequency variability is observed in the rotating case, likely due to large-scale structures generated during the filling process in rotation.

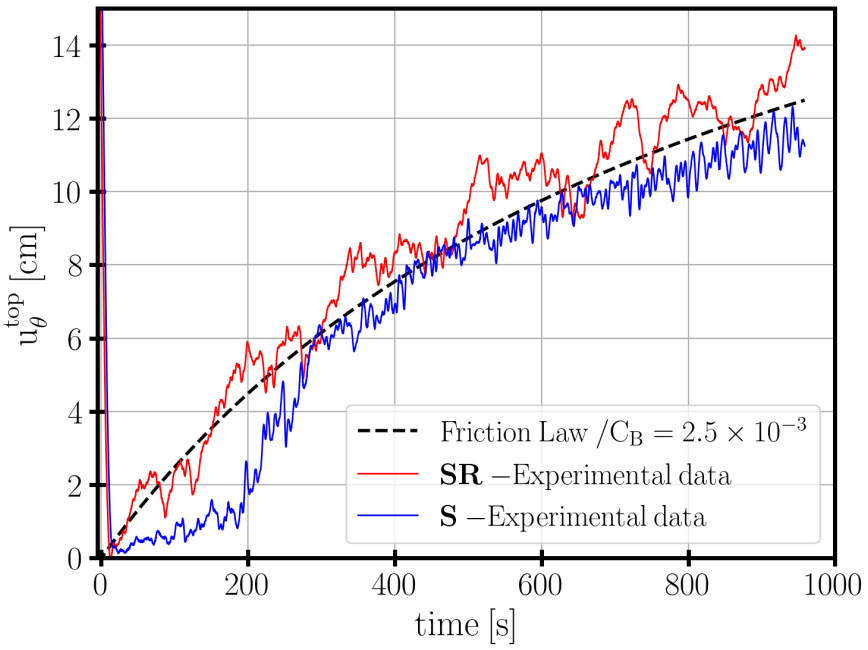


Figure 16. Time series of interior velocity for Spin-up case with (in red) and without (in blue) initial background rotation. The dashed black line correspond to the friction law given in Eq. (35). The interior velocity $\langle u_\theta^{top} \rangle$ is defined by average in the interval z=[13 13.5] cm, at the upper boundary of the observation window.

In the non-rotating case, we observe in Fig. 16 a first regime with virtually no change from $t = 20 - 200$s. In this initial stage, there is no spin-up process. The boundary layer only slowly grows by turbulent diffusion until it reaches the point of measurement, at $z = 13$ cm. This results in a sudden velocity increase at this location. At later times, some angular momentum is communicated to the bulk, and we end up with an evolution similar to the rotating case. Notably, this late friction law is very similar to the rotating case (EXP-SR).

The k-ε model provides a prediction to this spin-up process using Eq. (33). The model is strictly valid in the initial phase for which the bulk velocity is still small and assumed uniform across the fluid layer. The model predicts a friction force which quickly reaches a constant $u_*^2 \cos(\alpha_0) = 1.17$ cm^2 s^{-2}, while the term of radial momentum flux reaches a value $(4/r_0) \int uv dz = 0.18$ cm^2 s^{-2}. This steady regime is reached at time $t \simeq 100$ s for which the bulk velocity is $\langle u_\theta^{top} \rangle \simeq 3$ cm/s. This yields the friction coefficient $C_b = 2.5 \times 10^{-3}$ after division by $(v_0 - \langle u_\theta^{top} \rangle)^2$, which is consistent with the fitted value.

Note that the classical theory of the neutral atmospheric boundary layer yields the result (Garratt (1992), Sous et al. (2013)

$$\frac{u_*^2}{v^2} = \frac{\kappa^2}{(\ln Ro_* - A)^2 + B^2} \quad (36)$$

where $\kappa \simeq 0.41$ is the Von Karman constant, $A \simeq 1.3$ and $B \simeq 4.4$ are constants which characterise the shape of the Ekman layer velocity profile away from the log layer, and $Ro_* = u_*/(fz_0)$ is the friction Rossby number. In our case of smooth wall, we have $z_0 \simeq 0.12\nu/u_*$, so that $Ro_* = Re_*/0.12$, where Re_* was defined by Eq. 7. This relation yields $u_*^2/v^2 = 2.1 \times 10^{-3}$ which underestimates the measured friction by about 20 %. This difference, probably due to our moderate Reynolds number, is in agreement with the previous experimental results of Sous et al. (2013).

5.2. *Evolution of the vertical velocity profile*

Figure 17a shows the temporal evolution of the displacement thickness δ_1 for two non-rotating cases: unstratified (EXP-REF) and weakly stratified (EXP-S). The growth is identical for both cases and follows a linear trend $\delta_1 \propto t$, as expected from momentum conservation in the absence of rotation (see Eq. 15). The fit of experimental data with this linear evolution yields $C_\delta = 10^{-3}$, which corresponds to a ratio $u_*/v_0 \sim 1/30$. This value is smaller than the friction coefficient $C_b \simeq 2.5\ 10^{-3}$ previously obtained for the bulk spin-up over longer times. The value of C_δ is also about twice smaller than the corresponding result of Kozul et al. (2016) from DNS of a temporally evolving turbulent boundary layer. This reflects the fact that both friction and turbulence require some time to become fully established after the impulsive forcing. Based on the experimental curves, this transition phase lasts approximately 20 s (the forcing ramp itself is in the interval $t = [-10; 10]s$). By contrast an initial turbulence level is introduced in the DNS of Kozul et al. (2016), reproducing the effect of a trip wire often used to trigger turbulence in boundary layer studies.

The k-ε model reproduces the experimental growth of the displacement thickness reasonably well. During the first ~ 40 s, the agreement is particularly good, apart from the initial adjustment associated with the establishment of turbulence following the impulsive forcing. At later times, the model continues to reproduce the growth rate observed experimentally, although it systematically overestimates δ_1. The saturation observed in both the experiment and the model at late times ($t \gtrsim 80$s) occurs when the boundary layer extends beyond the PIV window, which defines the upper limit of the integral. In addition, at these times the boundary layer begins to feel the effect of rotation, which tends to inhibit its growth, as observed in Fig. 18.

Note that δ_1 is one possible indicator of the boundary layer thickness. An alternative indicator is the height δ_{99} at which the mean velocity reaches 99% of the outer velocity. This characterises roughly the layer with active turbulence. According to the DNS of Kozul et al. (2016), $\delta_1/\delta_{99} = C_1 u_*/v_0$ with $C_1 \simeq 4$. This would yield $\delta_{99} \simeq 7.5\delta_1$ with our result $u_*/v_0 \sim 1/30$. We cannot reliably measure δ_{99} because of limited measurement precision, but we measure δ_{95} at which the mean velocity reaches 95% of the outer velocity. We find typically $\delta_{95} \simeq 5\delta_1$, in agreement with the experiments with a rotating disk of Imayama et al. (2014), who found $\delta_{95} = \delta_1/0.19$.

We also estimate the boundary-layer thickness from the dye concentration, defining its upper boundary as the location of the maximum vertical gradient of the mean concentration (Fig. 5). Before the impulsive rotation is initiated, the dye is injected near the bottom as a thin layer of water at 20 °C into warmer water at 22 °C. After an initial period of mixing, this produces a fluorescent layer approximately 5 cm thick. Once the spin-up begins, turbulence rapidly erases the associated density stratification, so that the dye behaves as a passive scalar. As

shown in Fig. 17, the layer thickness increases approximately linearly from 5 to 12.5 cm over 110 s. Beyond this time, the measurement is again limited by the height of the observation window. Comparing the thickness inferred from the dye concentration, δ_{dye}, with δ_1, we find $\delta_{dye} \approx 6\delta_1$.

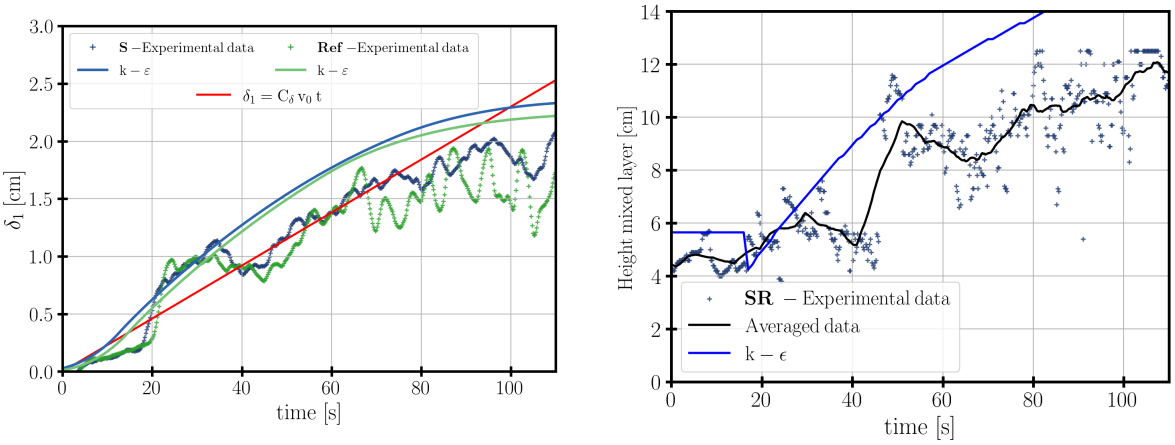


Figure 17. Left: Time evolution of the displacement thickness δ_1 (defined in Eq. 14). Two non-rotating configuration (EXP-Ref and EXP-S) are display with + markers and compared with their associated k-ε simulations (in solid line). The red line shows the linear fit for $\delta_1 \propto t$ with $C_\delta = 10^{-3}$, and $v_0 = 23.5$ cm s^{-1}. Right: Time evolution of the height of the mixed layer revealed by the maximum gradient of dye concentration for EXP-Ref. The thick black line is the average of the experimental data over a neighbour of 10 s.

5.3. *Rotation constraints on boundary layer depth*

In the rotating case (EXP-SR) the azimuthal velocity profile is not monotonic, with a maximum at height denoted $\delta_{U\,\max}$ (see Fig. 6). This height $\delta_{U\,\max}$ then provides a convenient characterisation of the boundary layer thickness. Figure 18a shows the temporal evolution of this thickness.

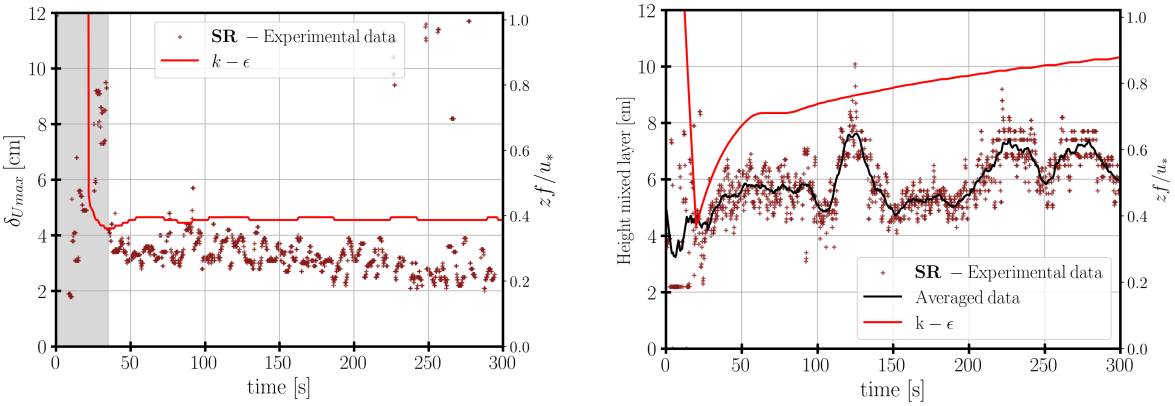


Figure 18. Left: Temporal evolution of the height of the maximum azimuthal velocity $\delta_{U_{max}}$. Right: same as figure 17b for experiment SR with rotation. For both figure, the experimental data marked with + and from the k-ε model (in solid line) and the adimensional scale on the right y-axis is given for a constant $u_* = v_0/20$ and a constant $f = 0.1\text{s}^{-1}$.

Following the step change in rotation rate at $t = 0$, the boundary layer develops from rest and $\delta_{U\,\max}$ grows rapidly. Within the gray shaded area, we observe a linear growth of $\delta_{U\,\max}$ for $0 \lesssim t \lesssim 30$ s: during this early phase, the flow is dominated by turbulent friction and particle trajectories are not yet significantly deflected by the Coriolis force. The large initial values of $\delta_{U\,\max}$ should not be interpreted as a physical overshoot: at early times, before

the Ekman spiral is established, the velocity maximum is very weak and the diagnostic is ill-defined. At $t \approx 30$ s ($\approx T_f/2$, i.e. half the inertial period) the Coriolis force becomes dynamically significant and the velocity profile develops a clear maximum, marking the onset of the Ekman spiral and the transition to a rotationally controlled regime. We see that $\delta_{U\,\max}$ then approaches the known Ekman depth $0.3\,u_*/f$. The result is in reasonable agreement with the k-ε model, which tends to a constant height $\delta_{U\,\max} = 0.4\,u_*/f$. This value is also in agreement with the DNS of Marlatt et al. (2012) (see Fig. 8).

At longer times, the experimental data show a gradual thinning of the boundary layer, unlike the model. This thinning is related to the spin-up process, which is not included in the model. Two effects are involved. Firstly, the shear stress u_* decreases as the velocity difference between the floor and the bulk decreases. Secondly, the spin-up increases the effective rotation rate of the fluid, increasing the effective Coriolis parameter f.

Figure 18b shows the corresponding growth of the boundary layer thickness characterised by the dye concentration. This is the counterpart of Figure 17b for a case with rotation. The horizontally inhomogeneous dye injection process does not allow us to observe the very initial growth phase unaffected by rotation. Oscillations observed at later times are attributed to the crossing of large horizontal eddies possibly produced during the fill-up of the tank or by detachment of the lateral boundary layers. Nevertheless, this figure enables us to demonstrate that the growth of the mixed layer is indeed limited by rotation (and not stratification). It initially stabilises to a value $h \simeq 0.5u_*/f$. The results from the k-ε model is somewhat larger, and a slow thickening persists on longer times. This is due to the diffusion of turbulence away from its shear driven production zone.

We do not observe the further decrease of the thickness observed for $\delta_{U\,\max}$. Indeed the layer of scalar concentration cannot 'un-mix'. Nevertheless, after 300 s, the height reached does not exceed 8 cm.

5.4. *Inertial oscillation in a turbulent regime*

We here characterize the inertial response of the Ekman layer using (i) time series at a fixed depth and (ii) depth-integrated transports, which are less sensitive to small-scale turbulent fluctuations.

5.4.1. *Temporal signature at a given depth*

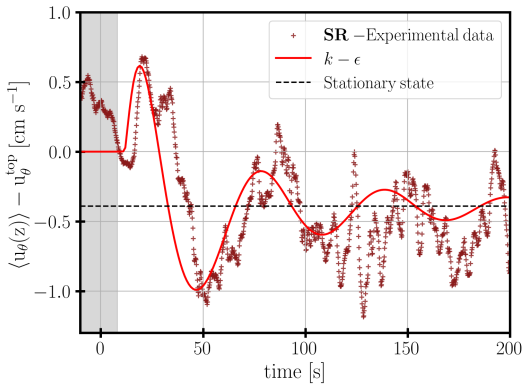


Figure 19. Dots: Temporal evolution of the defect velocity at $z = 3$ cm for EXP-SR. Solid line, same but for the k-ε model. The dashed black line indicates the stationary state after the damping of the oscillations in the k-ε model.

Figure 19 shows the evolution of the velocity defect ($\langle u_\theta \rangle - u_\theta^{\rm top}$) at a given altitude ($z = 3$ cm). We observe the onset of an oscillation at the inertial frequency f. The grey band marks the forcing ramp centred at $t = 0$. The oscillation starts around $t \simeq 10$ s, showing the

delay of turbulent diffusion from the plate to the altitude $z = 3$ cm. We observe that the velocity defect is not quite zero initially, reflecting residual velocities from the fill-up of the tank. Such large eddies are also responsible of random bursts at later times as they move through the measurement window. Yet damped inertial oscillations are clearly visible.

The 1D model reproduces well the first periods of oscillation, both in amplitude and phase, aside from a systematic phase shift of approximately 10 s. Agreement deteriorates beyond $t \approx 100$ s. Discrepancies are due to the intermittent bursts from large eddies, as mentioned above, but also to a persistence of the oscillations beyond the decay phase. This persistence may be due to a genuine mechanism of generation at resonance with the inertial frequency, not included in the columnar model.

The value of the drag coefficient $C_f = 5 \times 10^{-3}$ controls the damping rate of the inertial oscillations in the model: a larger C_f would extract momentum more efficiently from the oscillating flow and shorten the decay time. The agreement between the experimental data and the 1D model is an independent confirmation of the value of C_f.

5.4.2. *Temporal signature on horizontal transport*

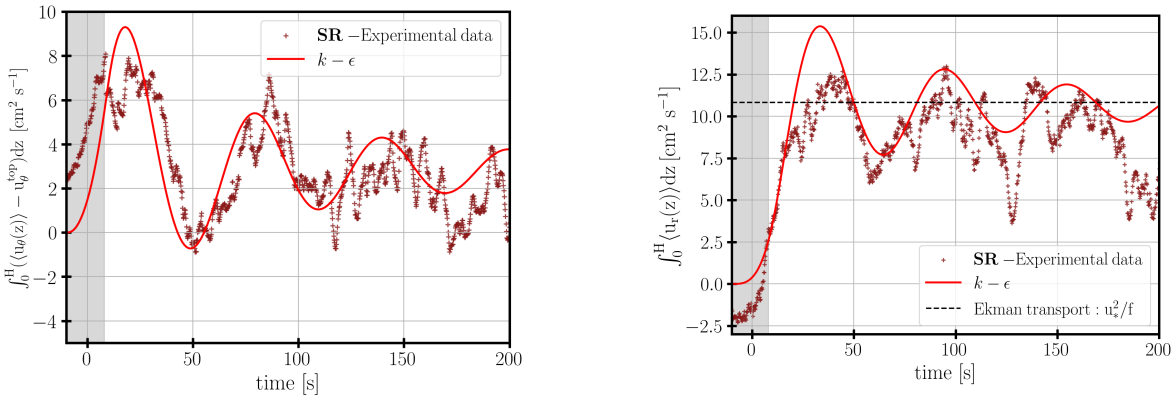


Figure 20. The temporal evolution of the transport between the bottom and $z = 6$ cm for the defect azimuthal velocity (left panel) and for the radial velocity (right panel). In both panel , the solid line show the k-ε model. The dashed black line show the theoretical Ekman transport u_*^2/f.

A more robust evaluation of this transient regime can be obtained by considering the vertically integrated transport over the height of the boundary layer (Fig. 20). This height, estimated at 6 cm in Fig. 7 and Fig. 6, remains relatively constant over the first 200 seconds in the case EXP-SR. Figure 20a (resp. 20b) shows the integrated transport between the wall and $z = 6$ cm for the defect azimuthal component (resp. for the radial component).

For the azimuthal defect transport (Figure 20a), the inertial oscillation appears more clearly than at a fixed height. We observe an oscillation close to the frequency f, with a decreasing amplitude. Once again, there is good agreement with the 1D model in terms of both amplitude and phase, although the same phase shift of the order of 10 s is observed.

The inertial oscillations are also apparent in the radial transport (Fig. 20b), with a phase shift of a quarter period as expected. They are however subjects to perturbations which appear to be uniform across the full height of the PIV window (probably due to large eddies). These perturbations are less apparent in the defect azimuthal component shown in Figure (Figure 20a), because of the subtraction by the velocity u_θ^{top}, which suppress the vertically uniform component.

6. Conclusions

In this study, we have investigated the transient dynamics of a fully turbulent Ekman boundary layer through large-scale spin-up experiments conducted on the Coriolis rotating platform in Grenoble. An important result is the reliable estimation of the friction velocity in a transient regime through several independent methods: extrapolation of the turbulent momentum flux profile toward the wall, fitting the interior velocity decay to the quadratic friction law (35), and direct comparison with the 1D model. This multi-method approach complements the work of Sous et al. (2013), who estimated the friction velocity under the assumption of a fully established regime

We also show that unlike the laminar case, where the attenuation rate is set analytically by the viscous Ekman layer, the turbulent experiments show that oscillations persist over several inertial periods, with an amplitude decay slower than predicted by laminar theory. The turbulent experimental case lies between the laminar no-slip and free-slip limits, a behaviour that is directly controlled by the type of bottom stress formulation. We show that the physically consistent representation of the bottom stress in a turbulent regime, where the viscous sublayer is not resolved, is neither a strict no-slip nor a free-slip condition, but rather a quadratic drag law of the form $\tau_b = C_f|\mathbf{u}|\mathbf{u}$. In the present range of parameters, calibration against DNS at comparable Reynolds number gives $C_f \approx 5 \times 10^{-3}$, whereas the logarithmic wall law yields a somewhat larger value, $C_f \approx 8.6 \times 10^{-3}$ (see Appendix A). This difference is likely related to the moderate Reynolds number of the experiments, for which the logarithmic layer is only marginally established. The fitted value of C_f should therefore be regarded as representative of the present regime rather than as universal. The precise value of C_f may vary with Reynolds number and, potentially, with Rossby number.

Furthermore, despite the slow attenuation of the oscillations, the experimental and numerical transports remain consistent with an evolution towards the steady Ekman transport u_*^2/f. Additional simulations (see Appendix B) with a much longer acceleration ramp, which strongly suppresses the excitation of inertial oscillations, converge towards the same final Ekman state as the impulsive case. This indicates that the steady Ekman state is independent of the transient inertial oscillations argues in favour of the linear superposition.

The k-ε closure is shown to faithfully reproduce the mean velocity profiles, the turbulent kinetic energy, and the Reynolds stress profiles of the turbulent Ekman layer. In particular, the model correctly captures the vertical confinement of the boundary layer by rotation, the amplitude and phase of the inertial oscillations during the first few periods, and the deepening of the mixed layer in the stratified cases. Estimates of the mixing length derived from the measured turbulent correlations, both with and without background rotation, are consistent with the k-ε scaling.

The k-ε closure relies on the Boussinesq eddy-viscosity hypothesis, which assumes that the Reynolds stress tensor is proportional to the mean strain-rate tensor through a single scalar isotropic eddy viscosity ν_t. In a standard shear-driven boundary layer, this assumption is a reasonable first approximation, and the k-ε model has been shown to reproduce mean velocity profiles and TKE budgets with acceptable accuracy (Pope 2000). However, in a rotating boundary layer such as the Ekman layer, the Coriolis force introduces a preferred direction that breaks the symmetry between the streamwise and cross-stream components of the Reynolds stress tensor (Svensson 1979). In particular, Direct Numerical Simulations (DNS) studies of the turbulent Ekman layer (Marlatt et al. 2012) have shown that the off-diagonal component $\langle u'v' \rangle$ makes a non-negligible contribution to the turbulent energy budget, a direct consequence of the rotation-induced coupling between the two horizontal velocity components (Wirth 2010). This anisotropy is not captured by the standard k-ε model, which uses a single eddy viscosity for all stress components. More sophisticated closures, such as Reynolds Stress Models (RSM), transport individual components of the Reynolds stress tensor and can, in principle, represent

this directional asymmetry (see e.g. Launder et al. 1975). In the present study, the good agreement between the k-ε model and the experimental data suggests that, at the Reynolds numbers considered, the isotropic approximation remains adequate for the mean flow and TKE, even if it ignores the cross-stream stress.

The present results have direct implications for the parameterization of oceanic boundary layers in numerical models. In coarse-resolution models, the grid spacing exceeds the thickness of the turbulent boundary layer, so that the bottom boundary condition is not merely a numerical detail but a physical one with large-scale consequences. The choice of bottom friction parameterization directly controls the energy dissipation rate of the large-scale flow (Gallet and Ferrari 2020, Thompson and Young 2006). This sensitivity underscores the importance of using a physically motivated drag law rather than an ad hoc boundary condition. Imposing a classical no-slip condition directly on the resolved velocity field is inconsistent with the law of the wall at coarse resolution: the molecular processes responsible for arresting the flow at the wall are not represented by the eddy-viscosity parameterizations used in such models, and the resulting dissipation is dominated by the artificial shear imposed to meet the no-slip condition (Deremble et al. 2011). Conversely, a free-slip condition sets the tangential stress to zero, which is equally unphysical for a solid boundary. The physically consistent approach is therefore to supplement a free-slip condition with a drag parameterization of the form $\tau_b = C_f|\mathbf{u}|\mathbf{u}$, in the spirit of the law of the wall, with $C_f \approx \mathcal{O}(10^{-3})$ as inferred from this study. The laboratory experiments considered here remain at Reynolds numbers several orders of magnitude below oceanic values, and the calibrated value of C_f should therefore not be extrapolated directly to geophysical conditions. At larger Reynolds numbers the wall-law estimate should provide a more appropriate reference. Future work should thus explore the dependence of C_f on Reynolds and Rossby numbers, as well as extend the present results to strongly stratified regimes where buoyancy limits the vertical penetration of momentum and the interaction between the Ekman layer and the pycnocline becomes dynamically important.

Acknowledgments

The GOTM model is available at `https://gotm.net/`. The UVMAT software is available at `https://gricad-gitlab.univ-grenoble-alpes.fr/legi/soft/uvmat-doc`

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Data availability statement

The data used in this article is available on Zenodo at Coppin et al. (2026).

Disclosure statement

The authors report there are no competing interests to declare.

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Appendix A: Single-column model setting: k-ε closure

We use the open-source GOTM model (General Ocean Turbulence Model) in the single column configuration to numerically solve Equations (13). This model solves the momentum, salt and heat transport equations in a 1D water column, using the k-ε model of turbulence. In this model, we add to the molecular viscosity and diffusivity an eddy viscosity ν_t and eddy diffusivity of buoyancy ν_t', which are locally expressed from the turbulent kinetic energy (TKE) k and the TKE dissipation rate ε as

$$\nu_t = c_\mu \frac{k^2}{\varepsilon} \qquad \nu_t' = c_\mu' \frac{k^2}{\varepsilon} \,. \text{(A.1)}$$

The coefficients c_μ and c_μ' depend on the local stratification and vertical shear. We use the formulation of Canuto et al. (2001) corresponding to the case denoted 'CA' by Burchard and Bolding (2001), and 'CHCD01A' by Umlauf and Burchard (2005). However the stratification effects remain small in our study, so the coefficients c_μ and c_μ' remain close to the value 0.09 classically used in shear flows with uniform density.

The equation for the TKE $k \equiv \overline{\mathbf{u}'^2}/2$ is written as a transport equation

$$\frac{\partial k}{\partial t} + \frac{\partial T_k}{\partial z} = P + B - \varepsilon \text{(A.2)}$$

involving an eddy flux T_k, a production rate P, a conversion rate to potential energy B, and the dissipation rate ε. P and B are exactly expressed from the correlations of turbulent fluctuations, while the eddy flux is modelled by a diffusion law $T_k = -\nu_t \partial k/\partial z$,

The production P is set by the shear according to

$$P = -\overline{u'w'}\frac{\partial u}{\partial z} - \overline{v'w'}\frac{\partial v}{\partial z} = \nu_t \left|\frac{\partial \mathbf{u}}{\partial z}\right|^2 \text{(A.3)}$$

(the notation $\bar{\cdot}$ and $\cdot'$ denote respectively the ensemble average and fluctuations from this average).

The conversion rate to potential energy B also represents the vertical flux of buoyancy. Within the eddy diffusivity hypothesis, it is expressed as

$$B = \overline{w'b'} = -\nu_t' \frac{\partial b}{\partial z} \equiv -\nu_t' N^2 \,, \text{(A.4)}$$

introducing the local buoyancy frequency $N(z,t)$.

A transport equation with similar form is empirically introduced for ε to close the system,

$$\frac{\partial \varepsilon}{\partial t} + \frac{\partial}{\partial z}\left(\frac{\nu_t}{\sigma_\varepsilon}\frac{\partial \varepsilon}{\partial z}\right) = \frac{\varepsilon}{k}\left(c_{\varepsilon 1}\, P + c_{\varepsilon 3}\, B - c_{\varepsilon 2}\, \varepsilon\right) \text{(A.5)}$$

We use the classical values $\sigma_\varepsilon = 1.3$, $c_{\varepsilon 1} = 1.44$, and $c_{\varepsilon 2} = 1.92$ (Rodi 1987). The parameter $c_{\varepsilon 3}$ is less consensual, but it is of weak relevance in our case of weak stratification. We use here $c_{\varepsilon 3} = -0.621$ when $B < 0$ and $c_{\varepsilon 3} = 1.5$ when $B > 0$ (Burchard and Bolding 2001). We added the terms due to rotation in Eq. (13) in the mean flow equations. Note that no molecular viscosity is added to the equations of TKE and ε.

The simulations are configured to match the laboratory experiments at radius $r_0 = 4.5$ m (the PIV measurement location). The vertical grid spacing is 0.1 cm over a domain extending 50 cm along the z-axis. Simulations run for 600 s with a time step of 0.1 s. We set the Coriolis parameter to $f = 0.105\ \mathrm{s}^{-1}$, corresponding to the initial rotation rate of the laboratory experiments, and set $f = 0$ in the non-rotating cases. The initial temperature profile used in the 1D

simulations are interpolated on the vertical grid of the model from experimental measurement. The surface stress is imposed as a linear ramp rather than an impulsive condition, in order to reproduce the finite spin-up time of the rotating platform. The origin $t = 0$ is defined as the midpoint of the ramp, where the velocity reaches half its nominal value. The influence of this ramp on the amplitude and phase of the inertial oscillation is studied in appendix B.

The boundary condition at $z = 0$ is a key difficulty. At the air-sea interface, the boundary condition is generally set as a given wind stress u_*^2 at the free surface. By contrast the wall velocity is imposed in our experiments, a common situation in the engineering context. Then the turbulent region must be matched with the viscous sub-layer, which requires appropriate modification of the k-ε or the related k-ω model (Speziale et al. 1992).

We here use a simplified approach by considering that our first grid point, at $z = z_1$ is in the log layer. This log-layer is characterized by a constant shear stress u_*^2 with an angle $-\alpha_0$ (outward with $\alpha_0 > 0$) with respect to the azimuthal direction. In this log layer, the fluid is well mixed, so that $B = 0$ in Eq. (A.2), and the TKE production locally balances dissipation. Considering that $\nu_t = \kappa u_* z$ (where $\kappa \simeq 0.41$ is the von Karman constant), we get for $z \to 0$,

$$\kappa z \frac{\partial u}{\partial z} \to u_* \sin\alpha_0 \quad \text{(A.6a)}$$

$$\kappa z \frac{\partial v}{\partial z} \to -u_* \cos\alpha_0 \quad \text{(A.6b)}$$

$$k = c_\mu^{-1/2} u_*^2 \quad \text{(A.6c)}$$

$$\varepsilon = P = u_*^3/(\kappa z) \quad \text{(A.6d)}$$

This provides an appropriate set of boundary conditions for the k-ε equations if the wall stress is imposed. Note that in the set of equations A.6, we have neglected molecular viscosity, which is a standard assumption in this derivation. It has been shown by Burchard and Baumert (1998) that Neumann conditions are numerically more stable than the Dirichlet conditions (A.6c-d). We obtain the Neumann boundary conditions by differentiating Eqs. (A.6c-d) with respect to z:

$$\frac{\partial k}{\partial z} = 0, \quad \text{(A.7a)}$$

$$\frac{\partial \varepsilon}{\partial z} = -\frac{u_*^3}{\kappa (z_1)^2} . \quad \text{(A.7b)}$$

The no-flux condition is also used for the temperature.

From (A.6a) and (A.6b), the velocity diverges logarithmically with the constant veering angle α_0,

$$u(z) = \frac{u_*}{\kappa} \sin\alpha_0 \ \ln\left(\frac{z}{z_0}\right) \quad \text{(A.8a)}$$

$$v(z) - v_0 = -\frac{u_*}{\kappa} \cos\alpha_0 \ \ln\left(\frac{z}{z_0}\right) \quad \text{(A.8b)}$$

This is not a problem in the case of an imposed stress, as the dynamics in the turbulent layer is fully governed by the shear stress u_*. This contrasts with the case of an imposed velocity, for which the stress must be related to the wall velocity. Near a smooth wall, the classical expression $|\mathbf{u} - \mathbf{u_{wall}}|/u_* = \kappa^{-1} \ln(z u_*/\nu) + 5.2$ generally holds (see for instance Pope 2000). This is recovered from (A.8) with the relation $z_0 = (\nu/u_*) \ \exp(-5.2\ \kappa) \simeq \ 0.12\ \nu/u_*$.

Let us assume that the first grid point z_1 is in the log layer, so that $z_1 \gg z_0$. There is a velocity shift $\mathbf{u}(z_1) - \mathbf{u_{wall}}$ with respect to the boundary velocity $\mathbf{u_{wall}}$ which is aligned with the shear stress. The square of the modulus of this velocity shift is proportional to the shear stress u_*^2,

$$u_*^2 = C_f |\mathbf{u}(z_1) - \mathbf{u_{wall}}|^2 \quad \text{(A.9)}$$

Comparison with Eq. (A.8) yields

$$C_f = \left[\frac{\kappa}{\ln(z_1/z_0)} \right]^2 \quad \text{(A.10)}$$

With the expression $z_0 = 0.12\, \nu/u_*$, this implicit relation fully defines the boundary condition for a given wall velocity $\mathbf{u_{wall}}$. It could be extended to a rough surface, by expressing z_0 in terms of a roughness height instead of ν/u_*, which is generally more appropriate for geophysical flows.

This expression applies within the layer of constant stress, which is limited to an altitude estimated as $z_2 \simeq 0.02 u_*/f$ in the turbulent Ekman layer (see e.g. Garrett et al. (1993). Therefore, z_1 must satisfy the conditions $z_1 < z_2$. The ratio of these two bounds is $z_2/z_0 \simeq 0.2\, Re_*$ for a smooth wall, where Re_* is the Reynolds number defined by Eq. (7).

In our experiments u_* remains of the order of 1 cm s^{-1} and $f \simeq 0.1$ s^{-1}, so that $z_2 \simeq 2$ mm. We choose $z_1 = 1$ mm as the first model point to remain well in the constant shear zone, even in unsteady regimes. Then the previous expression based on the log layer gives $C_f = 8.6 \times 10^{-3}$. For these parameters, the first model grid point corresponds to $y^+ = z_1 u_*/\nu \sim 10$. This is in the so-called buffer layer, while the log layer is generally considered to begin at $y^+ \simeq 30$. A log layer cannot really develop for $Re_* \simeq 1000$. For $y^+ = 10$, classical results on the buffer zone indicate that about half of the shear stress is due to turbulence, and the other half to viscosity. This leads us to choose a lower value of C_f.

Further adjustment is made by comparison with the DNS of Marlatt et al. (2012), leading to the value $C_f = 5 \times 10^{-3}$. The drag coefficient C_f is calibrated by matching the steady-state cross-isobaric angle α_0, a well established reference value. Our k-ε simulations are run for the pure Ekman layer (without centrifugal acceleration terms), and are integrated until inertial oscillations are fully damped. Figure A1 shows α_0 as a function of C_f. For $z_1 = 0.1$ cm, the DNS reference value $\alpha_0^{\mathrm{DNS}} = 18.56°$ is matched at $C_f \approx 5 \times 10^{-3}$, which is therefore adopted in all the k-ε simulations presented in this study. This choice also provides an optimum match to the experimental data (see Fig. 8). Increasing this friction parameter increases the velocity at $z = z_1$ and increases u_* and the boundary layer thickness, in accordance to the scaling in u_*/f (while keeping similar velocity profiles after rescaling). We find however that the global results are not very sensitive to this parameter choice.

Appendix B: Influence of a varying forcing ramp

We carried out k-ε simulations with different ramp durations (0, 20, 30, 40 and 60 s). These simulations show how the ramp duration primarily affects the initial amplitude and phase of the inertial oscillations, whilst the final equilibrium remains unchanged (see Fig. B1). When the ramp is widened, the oscillations are delayed and their amplitude decreases, but their decay remains broadly within the same envelope. This suggests that the damping rate is primarily controlled by the drag coefficient C_f rather than by the precise form of the initial forcing.

We therefore regard the impulsive initial condition as a useful approximation for theoretical analysis, whilst the finite ramp is taken into account in the simulations used for comparison with the experiment. With our ramp duration of 20 s, the difference with respect to the

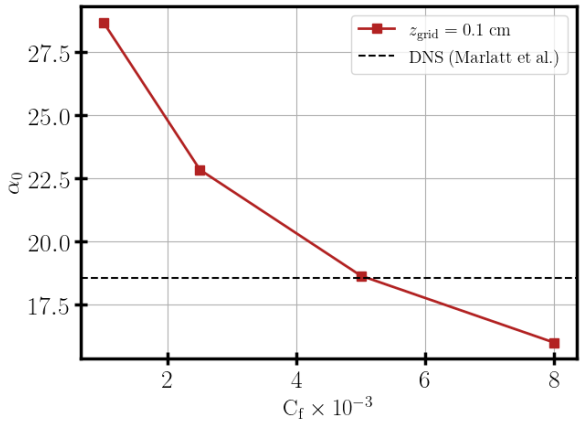


Figure A1. Cross-isobar angle measured a the first grid point as a function of the bottom drag coefficient C_f, computed at steady state (inertial oscillation are fully damped) using the k-ε model. The horizontal line indicates the DNS reference values of reported by Marlatt et al. (2012) (18.56, dashed) for a steady Ekman layer at $Re = 1000$. Centrifugal acceleration terms are excluded in this comparison.

impulsive start remains small.

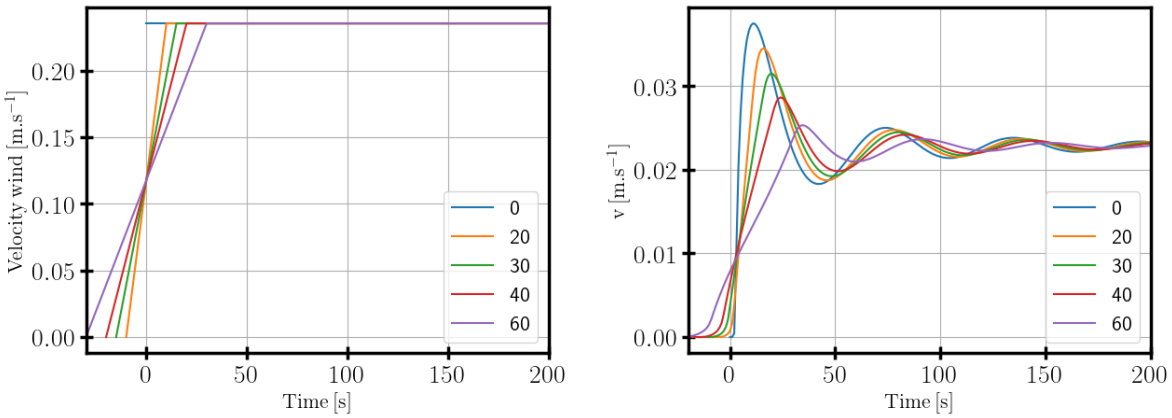


Figure B1. Time evolution of: (left) impose wind velocity; (right) the velocity v aligned with the wind stress at 1 cm above the plate. Different simulation are displayed with various forcing ramp acceleration (0 s, 20 s,30 s, 40 s and 60 s). The ramp use to reproduce the Coriolis platform acceleration is 20 s. The time origin is set at the middle of the ramp.